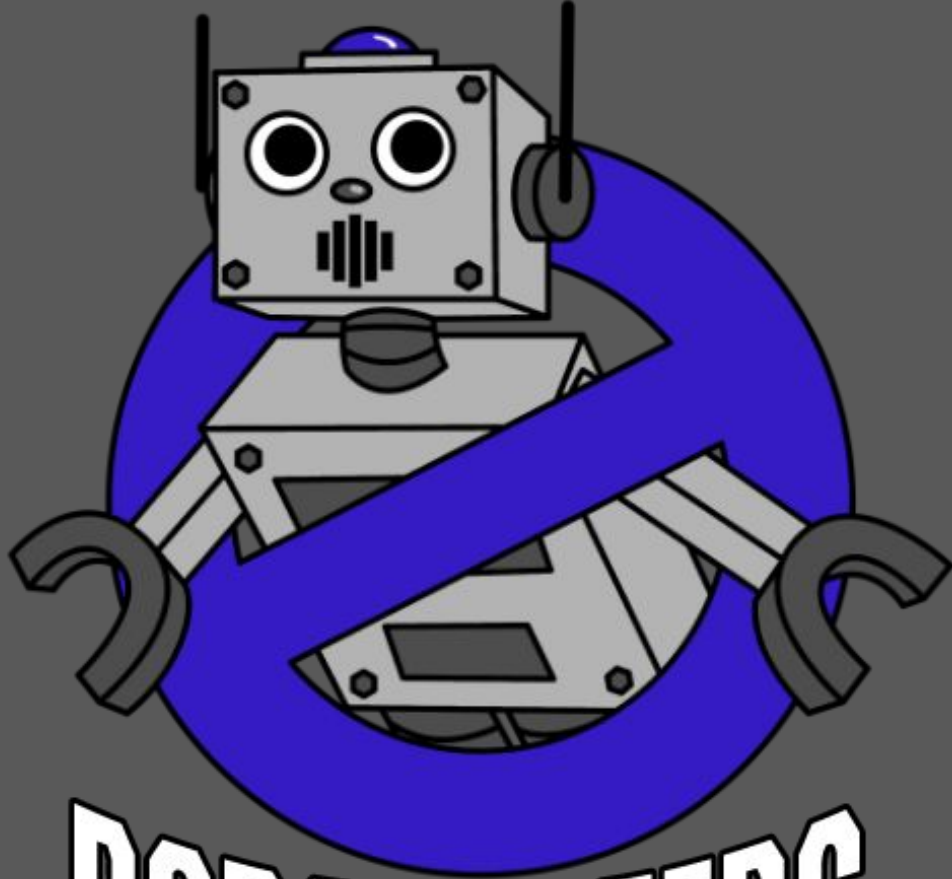


597C



ROBOBUSTERS

RoboBusters Blue

SEM

Robotics Engineering
Notebook

Challenge: Push Back

Start Date: 09/01/2025

End Date: 3/8/2026

Table of Contents

Page #	Title	Date
6	Meet the Team	September 6
7-14	Planning Calendar	
15	Section A - CAD & Construction	
74	Section B - Programming	
16	Analyzing the Game	
17	Analyzing the Game (Cont.)	
18	Field Elements	
19	Field Elements (Cont.)	
20	Design Analysis	
21	Brainstorming: Game Strategy	
22	Brainstorming: Game Strategy (Cont.)	
23	Planning the Game Strategy	
24-25	Planning the Game Strategy (Cont.)	
26	Finalizing the Game Strategy	
27	Brainstorming	September 8
28	Brainstorming Structure	
29-34	Brainstorming Structure (Cont.)	
35	Design Conclusions	
36	Building the Drivetrain	September 16
37	Improving the Drivetrain	September 20
38	Building the Intake & Elevator	September 23

Table of Contents

Page #	Title	Date
39	Building the Intake & Elevator (Cont.)	September 23
40	Building the Cage	September 29
41	Designing the Dispenser	October 1
42	Dispenser Prototyping	↓
43	Dispenser Prototyping (Cont.)	October 4
44	Improving the Cage	October 28
45	Improving the Intake	↓
46	Trenton Tournament Nov.1 - Notes and Reflection	November 1
47	Modifying the Outtake	November 5
48	Phoenix Premiere Robotics Blended Tournament Nov.8 - Notes and Reflections	November 8
49	Pneumatics Part 1 - Brainstorming	November 10
50	Pneumatics Part 2 - Adding Tanks	November 11
51	Pneumatics Part 3 - Descoring & Match Loader Mechanism	November 12
52	SSHS Robotics Rumble [Blended] Nov. 15 - Notes and Reflection	November 15
53	Miller Grove Tournament Nov. 21 - Notes and Reflection	November 21
54	Vex V5 Push Back Redesign Research	December 1
55	Video Analysis	↓
56	RuiGuan Subsystems Breakdown	↓
57	Start of robot rebuild	December 3
58	Flexible Intake Subsystem	January 2

Table of Contents

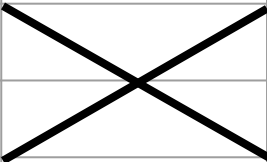
Page #	Title	Date
59	Ramp Elevator Subsystem	January 2
60	Outtake Flap Subsystem	January 5
61	Matchload Subsystem	↓
62	Trenton Winter Classic Tournament Jan.10 - Notes and Reflection	January 10
63	Outtake Flap Redesign	January 13
64	Aligner Redesign	↓
65	Ramp Funnel	↓
66	Snow Bots Miller Grove High School Tournament Jan.17 - Notes and Reflection	January 17
67	The Battle For Valhalla Tournament: School Based Tournament Jan.31 - Notes and Reflection	January 31
68	Dallas ISD Skyline's 9th Annual VRC Tournament - Notes and Reflection	February 14
69	Drivetrain Redesign	February 16
70	Drivetrain Redesign Cont.	↓
71	Ramp Elevator Improvement	February 17
72	Ramp Funnel Redesign	February 18
73	Matchload Redesign	February 25
74	Section B - Programming	
75	Button Map	
76	DATA NORMALIZATION	August 12, 2025
77	SYSTEM Initialization	↓

Table of Contents

Page #	Title	Date
78	CONSTANTS configuration	August 23, 2025
79	DRIVETRAIN: Initialization	↓
80	PID CONTROLLER	
81	DriveTrain: Set Velocity	
82	DriveTrain: Set Open Loop	↓
83	INTAKE SUBSYSTEM	
84	Programming: Constants File Setup	
85-87	Programming: DriveTrain Subsystem Implementation Part 1	↓
88	Programming: DriveTrain Subsystem Implementation Part 2	
89	Programming: Autonomous – Initial Pathing and Intake Part 1	
90	Programming: Autonomous – ball and goal alignment	↓
91	Programming: Autonomous – Initial Pathing and Intake Part 2	
92	DeScore	October 29, 2025
93	MATCHLOAD CYCLE	November 25, 2025
94	IMU WRAPPER	January, 6,2026
95	COLOR SORT LOGIC	January,16,2026
96	Absolute Position Turning	March 6, 2026
97	On-Screen Autonomous Selector	March 6, 2026
98	GitHub Repositories	

Meet the Team

September 6, 2025

Written by: J.C., D.M., A.B., T.C., D.S.

Present: J.C., D.M., A.B., T.C., D.S.

J Cruz (J.C.) - Team Captain

Yellow! Im J, a senior at SEM. I've been doing robotics for 5 years as a builder and sometimes driver. I am interested in engineering specifically in robotics and mechatronics.

Tyrese Cox (T.C.) - Programmer & Driver

I'm a senior at SEM. I've been doing robotics since 3rd grade and plan on becoming an aerospace engineer

Axel Banegas (A.B.) - CADder & Builder

I'm a sophomore at SEM. Since elementary, I've enjoyed engineering — whether with LEGO at home or with metal and screws at competitions. Now, I am especially interested in the design aspect of engineering.

David Moreno (D.M.) - Builder

I'm a freshman at Townview LAW. I've done robotics(vex iq, lego league) in middle school and I am very interested in robotics

Damian Sarmiento (D.S.) - Builder

I'm a junior at SEM. I've done robotics for a while now and I enjoy constructing robots.

Planning Calendar

September 11, 2025

Written by: A.B., J.C.

Present: A.B., J.C.

September Calendar

Building	Programming	Planning	Testing
----------	-------------	----------	---------

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1	2	3	4	5	6 Intros & Game Analysis
7	8 Brainstorming Designs	9	10	11 Calendar Work / Finalizing Designs	12	13
14	15 Programming Drivetrain	16 Starting Drivetrain	17	18	19	20 Programming Controller + Drivetrain
21	22	23 Intake & Elevator	24	25	26	27
28	29 Robot Cage	30				

Planning Calendar

September 11, 2025

Written by: A.B., J.C.

Present: A.B., J.C.

October Calendar

Building	Programming	Planning	Testing
----------	-------------	----------	---------

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
			1 Dispenser & Drive Testing	2	3	4 Testing Robot Functions
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20 Improving Program	21	22	23 	24	25
26	27 Improving the Cage and Intake Subsystems	28	29	30	31	

Planning Calendar

September 11, 2025

Written by: A.B., J.C.

Present: A.B., J.C.

November Calendar

Building	Programming	Planning	Testing
----------	-------------	----------	---------

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
						1 Comp Strats
2	3	4	5 Fixing Outtake	6	7	8 Comp Strats
9	10 Brainstorming + Adding Pneumatics	11	12	13	14	15
16	17 Debriefing Past Comps / Reflecting	18	19	20	21  Miller Grove Competition	22
23	24 December Calendar	25	26	27	28	29
30						

Planning Calendar

September 11, 2025

Written by: A.B., J.C.

Present: A.B., J.C.

December Calendar

Building	Programming	Planning	Testing
----------	-------------	----------	---------

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1 Comp Debrief	2	3 New Design Research	4	5	6
7	8 Design Research (Pt.2)	9	10 Ruiguan Bot Research	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
Christmas Break						
28	29	30	31 			

Planning Calendar

September 11, 2025

Written by: A.B, J.C.

Present: A.B., J.C.

January Calendar

Building	Programming	Planning	Testing
----------	-------------	----------	---------

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
				1	2 Bot Rebuild	3 Rebuild
4	5 Outtake Flap & Matchload	6	7	8	9	10 Trenton Competition Strats + Reflection
11	12	13 Outtake Flap Redesign	14	15	16	17 Miller Grove Competition
18	19	20	21	22	23	24 
25 	26 	27 	28 	29 	30	31 Irving Competition

Planning Calendar

September 11, 2025

Written by: A.B., J.C.

Present: A.B., J.C.

February Calendar

Building	Programming	Planning	Testing
----------	-------------	----------	---------

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1	2 No Meeting	3	4	5	6	7
8	9 No Meeting	10	11	12	13	14  Skyline Competition
15	16 Drivetrain Redesign	17 Ramp Elevator Revising	18	19	20	21
22	23	24	25	26	27	28
Work on Autonomous						

Planning Calendar

September 11, 2025

Written by: A.B., J.C.

Present: A.B., J.C.

March Calendar

Building	Programming	Planning	Testing
----------	-------------	----------	---------

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1	2 Work on Autonomous	3	4	5	6	7
8	9	10	11	12	13  STATE CHAMPIONSHIP	14 
15	16	17	18	19	20	21
22	23	24	25	26	27	28

Planning Calendar

September 11, 2025

Written by: A.B., J.C.

Present: A.B., J.C.

April Calendar

Building	Programming	Planning	Testing
-----------------	--------------------	-----------------	----------------

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28

RoboBusters Blue

SEM

**Robotics Engineering
Notebook**

**Section A - CAD &
Construction**

Analyzing the Game

September 6, 2025

Written by: A.B.

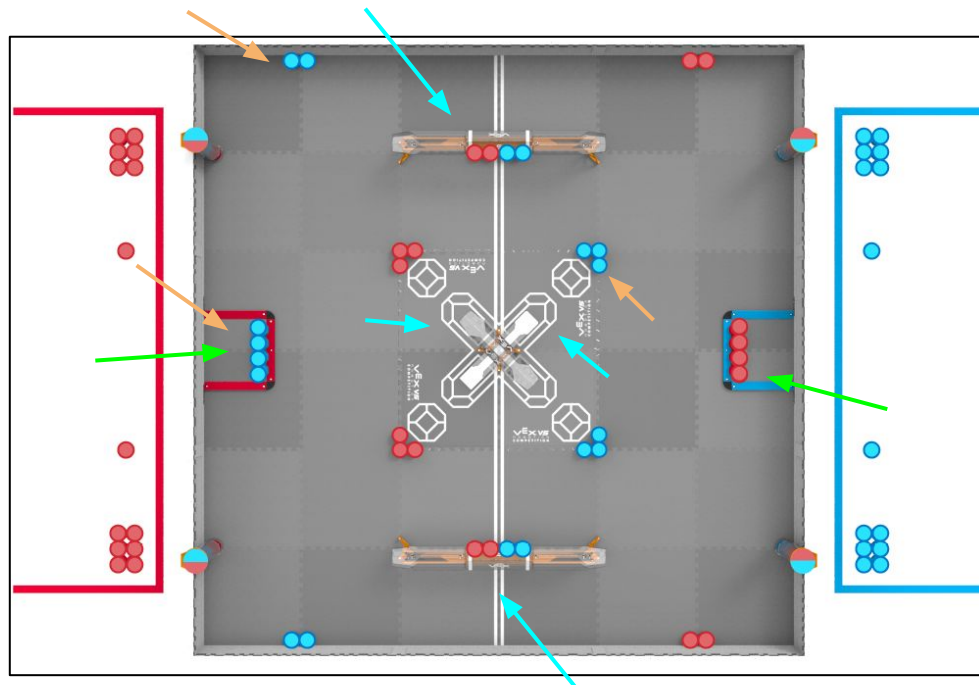
Present: J.C., D.M., A.B.

The Goal

Alliances must score as many blocks as possible inside the goals, while simultaneously attaining majority control in the center of the goals for bonus points. Furthermore, they must work together to complete additional bonus tasks such as those for the Autonomous Win Point and the Alliance Parking Bonus.

Scoring

- Every block inside of a goal equates to 3 points
- Obtaining center control of any outer goal equates to 10 points
- Obtaining control of the top middle goal equates to 8 points
- Obtaining control of the bottom middle goal equates to 6 points
- Placing *one* robot in the alliance parking equates to 8 points
- Placing *both* robots in the alliance parking equates to 30 points
- Scoring the most points during the autonomous period equates to 10 points



Blocks		Goals		Alliance Zone	
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Analyzing the Game (Cont.)

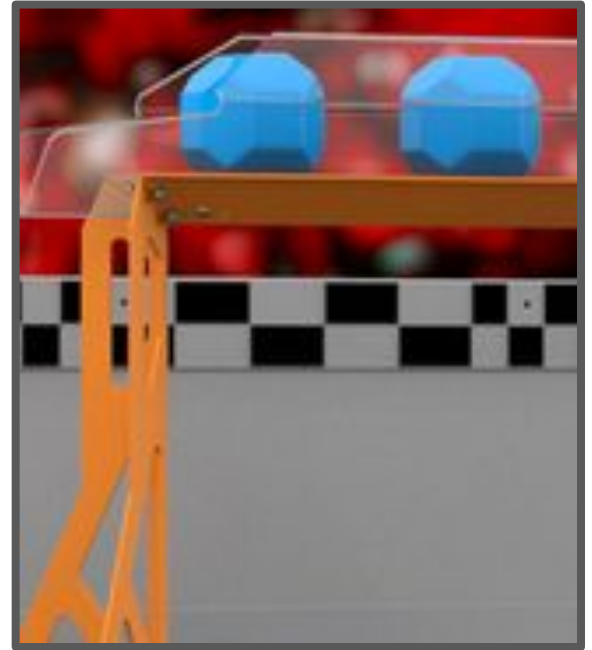
September 6, 2025

Written by: A.B.

Present: J.C., D.M., A.B.

Rules to Remember

- Alliances play on a 12 by 12 foot field
- The game is comprised of two alliances, each with two teams
- There's a total of 88 blocks, 44 blocks for each color
- There is *no* possession limit
- The game begins with a fifteen second autonomous period
- Only blocks *fully* inside the control zone count toward the control bonus



Challenges of the Game

Match unpredictability will probably make the coordination and parking of *both* alliance bots very difficult.

Robots with larger storage can afford to spend the majority of their time collecting blocks in order to quickly score them later on.

We need a minimum of two autonomous programs with different starting positions. This is to guarantee that we have a *usable* program if our alliance member needs a specific side of the field.

Scoring in itself will be a hard task if we are constantly being pushed around, making it imperative that we can stabilize our robot for a few seconds.

We can't have multiple robots, so we must decide on what type we want to have and stick with for the majority of the season.

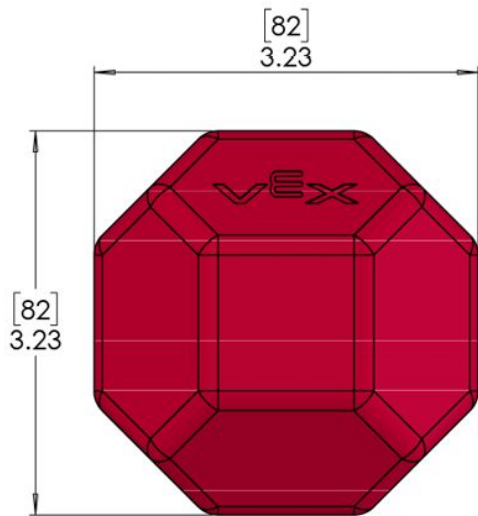
Field Elements

September 6, 2025

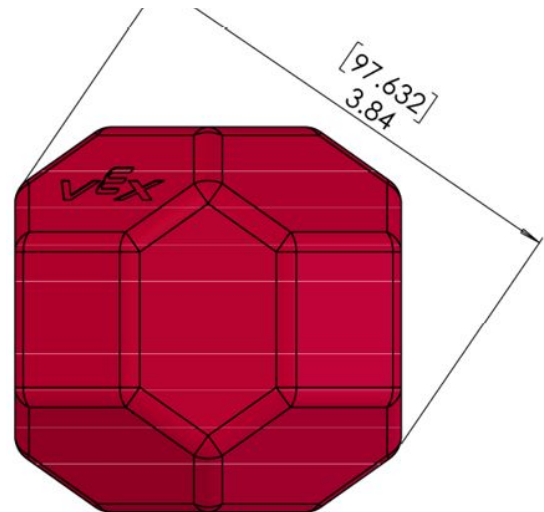
Written by: A.B.

Present: J.C., D.M., A.B.

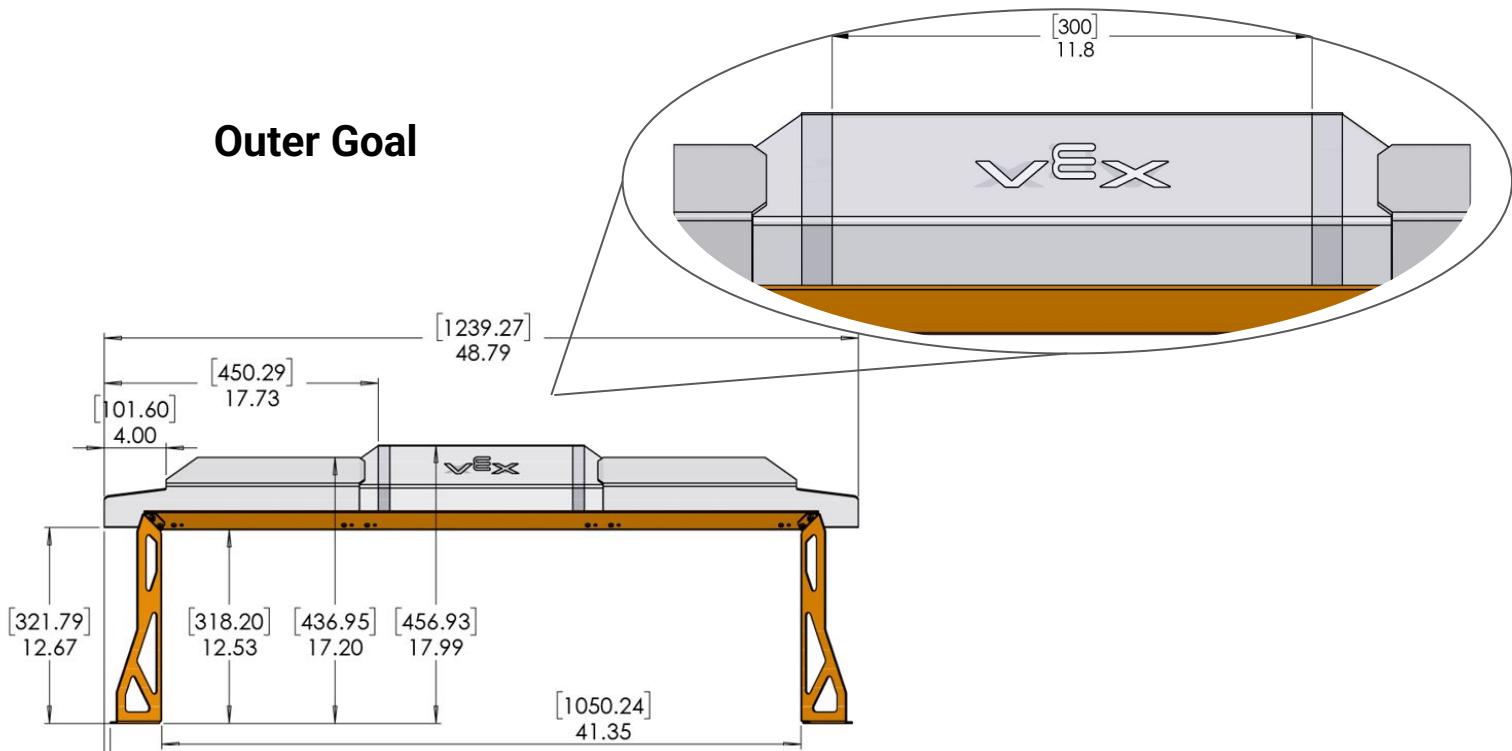
- Blocks measure 82 mm from face to face, and weigh 40 ± 4 grams
- Long goals (1239 mm) can hold 15 blocks
- Outer goal control zones (300 mm) can hold 3 blocks



Block



Outer Goal



Field Elements (Cont.)

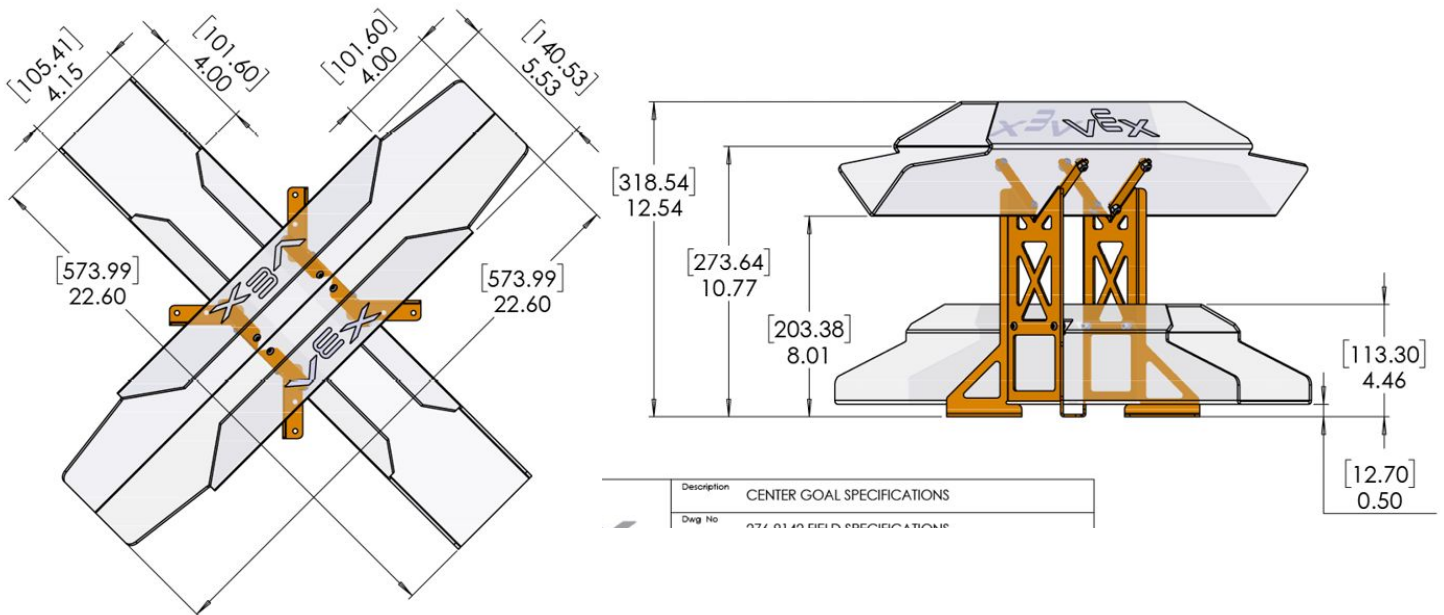
September 6, 2025

Written by: A.B.

Present: J.C., D.M., A.B.

- Center goals (both ~574 mm) can hold 7 blocks

Center Goals



A.B. 9/6/25

Design Analysis

September 6, 2025

Written by: A.B.

Present: J.C., D.M., A.B.

Design Goals/Challenges/Restrictions

- Size constraints
 - Robots are constrained to a base volume of 18*18*18 inches cubed
 - Expansion increases the volume limit to 22*22*22 inches cubed
- Max motors
 - We may use any combination of VEX V5 Smart Motors (11W) (276-4840) and Smart Motors (5.5W) (276-4842), however The combined power of all motors (11W & 5.5W) must not exceed 88W
- Max sensors
 - There is no limit to the amount of sensors that can be used, though there are limitations to their usage.
- Gear for speed/torque
 - Gearing for speed may give us an edge when collecting blocks, however gearing torque will allow our robot to hold its ground and thus score without much interference from opponents.
 - It is also going to be very difficult to reach top speeds with constant defense, effectively reducing its usefulness.
- Things to be successful in the challenge
 - Have a good center of gravity, in order to prevent unwanted tilting when driving the robot.
 - Add sufficient spacing within the robot to be able to hold as many blocks as possible.
 - Have structural integrity within our robot. Parts shouldn't be loose or moving freely, as this reduces our chances of scoring.

A.B. 9/6/25

Brainstorming: Game Strategy

September 6, 2025

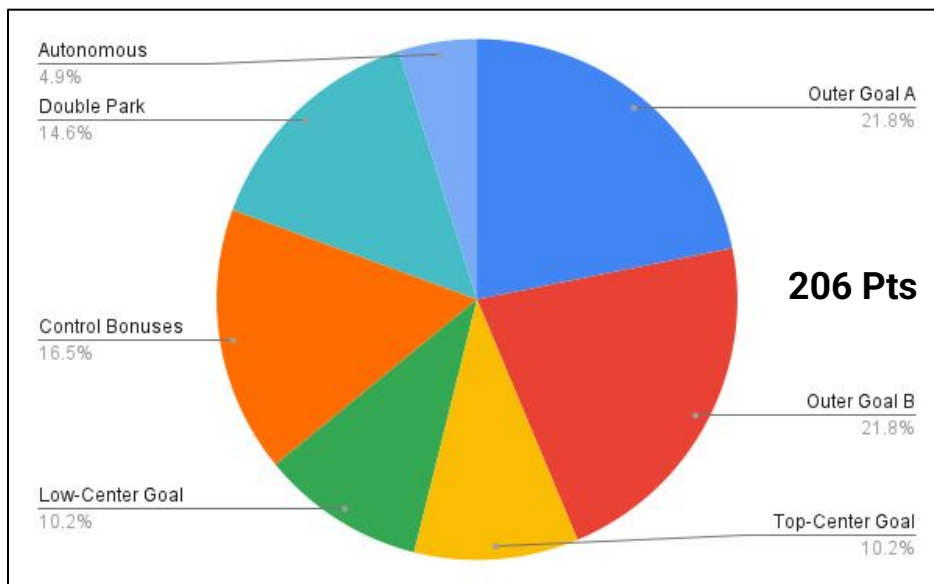
Written by: A.B.

Present: J.C., D.M., A.B.

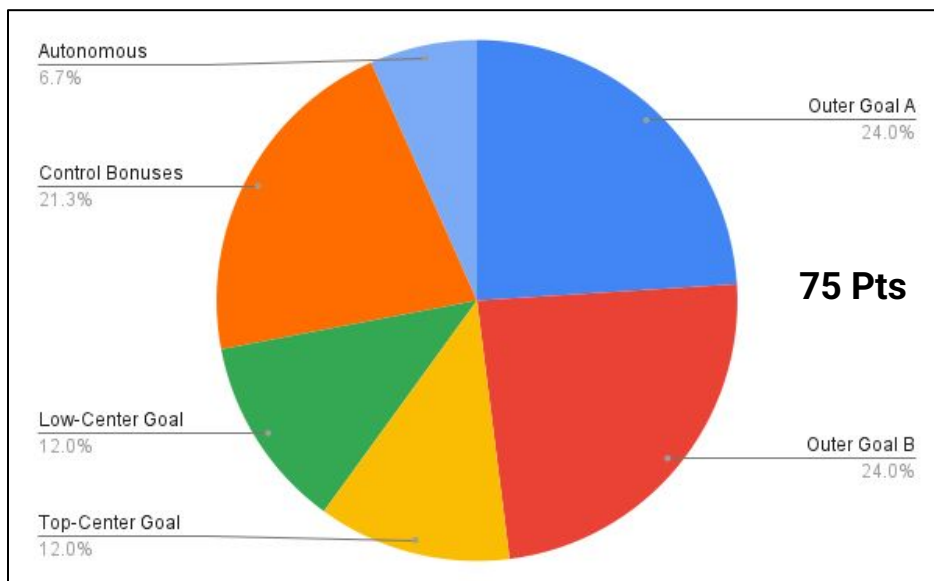
Goal

Come up with viable game plans that assume alliance variability and incorporate dynamic structures, rather than static ones.

Point Distribution



**Theoretical
Maximum**



**Expected
Performance**

Brainstorming: Game Strategy (Cont.)

September 6, 2025

Written by: A.B.

Present: J.C., D.M., A.B.

Reason for Analysis

- Analyzing ideal vs expected situations allows us to design a robot with the intent of scoring high amounts of points consistently, not with the intent of obtaining a perfect score.
- While scoring perfect scores is possible, it is not very plausible.
- Even with the perfect robot, we must also consider both our alliance variability and our opponent variability.
- Understanding this now reduces the chance of overbuilding, or focusing on the wrong strategies/designs early in the season.

Data Analysis

- For the theoretical maximum, we assumed perfect gameplay with zero opposition
- For the expected performance we made reasonable assumptions of likely gameplay
- Expected performance is only about 36.41% of the theoretical max
- Majority of the point potential lies in the outer goals
- Control Bonuses gained point potential in our expected performance
- On paper, the autonomous bonus points contribute the least to the point potential for match.
- Both of the charts show a generalized distribution of possible point scoring opportunities.
- While a section may have high scoring potential, we must also consider its reliability when selecting a strategy.

Planning the Game Strategy

September 6, 2025

Written by: A.B.

Present: J.C., D.M., A.B.

Do we Prioritize Autonomous?

Option 1: Yes

Pros	Cons
• Allows early scoring • Allows action without opponent interference	• Hard to perfect • Will need constant fixing if the robot is modified • Only rewards 10 points at most

Option 2: No

Pros	Cons
• Gives us more time to work on the robot's physical structure	• Gives up the chance for bonus points all together • Makes it easier for opponent to gain additional points

Notes:

- While 10 points is not a lot on its own, it is still useful to have guaranteed.
- Blocks can be knocked out, but the autonomous bonus can't
- It may be a bit time consuming, but the end result will definitely be worth it if we achieve consistency

Decision

While we won't be allocating *all* of our time to autonomous, we *will* extensively work on it in order to boost our score early on, or to give us a better strategic position before the driving period.

Planning the Game Strategy (Cont.)

September 6, 2025

Written by: A.B.

Present: J.C., D.M., A.B.

Focus on Outer or Center Goals?

Option 1: Focus only on Outer Goals

Pros	Cons
• Can store more blocks • Less opponent interference	• Harder to control • Permits opponents to score in the center

Option 2: Focus only on Center Goals

Pros	Cons
• Easy to control • Requires less blocks to fill	• Easier to know blocks off • Opponents can easily push us while attempting to score

Option 3: Focus on Both

Pros	Cons
• Allows for more flexibility • Prevents overcommitment • Opponents will have a harder time defending	• Final score is bound to be inconsistent • Applies more active decision pressure on driver

Notes:

- Focusing on one goal type reduces our flexibility
- Focusing on both goals requires a more complex bot

Decision

As a collective, we decided that we are willing to put in the work, if it means our robot has more capabilities and is less limited to specific scenarios.

Planning the Game Strategy (Cont.)

September 6, 2025

Written by: A.B.

Present: J.C., D.M., A.B.

Offense or Defense?

Option 1: Offense only

Pros	Cons
- Increases our point scoring - If done right, may lead to opponents being distracted on defense	- Opponents with better offense will ultimately win - Relies heavily on our teammate defending alone

Option 2: Defense only

Pros	Cons
- Easy to control - Requires less blocks to fill	- Easier to know blocks off - Opponents can easily push us while attempting to score

Option 3: Both Offense and Defense

Pros	Cons
- Allows for more flexibility - Prevents overcommitment - Opponents will have a harder time defending	May be harder to manage under pressure

Notes:

- Picking one specific style limits our performance under varying conditions

Decision

We want to score high as often as possible. Thus we need to be versatile and be capable of both offense and defense

Finalizing the Game Strategy

September 6, 2025

Written by: A.B.

Present: J.C., D.M., A.B.

Strategy Notes Overall:

- We don't yet *know* what the 'meta' will look like
- Neither does anyone else. Right now, everyone is experimenting
- Scoring reliability thus beats specialization
- If we wish to succeed, our robot needs to be strategically flexible
- Overcommitting into specific strategies leaves us vulnerable to the unpredictability of the matches
- It is currently safer to develop a strategy that doesn't fully rely on our alliance.

Strategy V1 (Conclusion)

In light of all of our brainstorming, we have decided to simply follow the basics of the game for our initial strategy.

At the start of the game, we plan to have a developed autonomous program that allows for the scoring of at least 2 blocks. While the autonomous win point *is* appealing, it is much more complicated to obtain than the autonomous bonus points. Therefore, for now, we will only focus on the autonomous bonus.

During the driving period, the plan is simple: obtain more points than our opponents. This essentially means we need to collect and score ample amounts of blocks, specifically concentrating them toward control zones whenever possible. At the same time, however, we must limit the point yield our opponents obtain. It will not be easy to do, but this means that we need to be aware of *all* of the robots on the field. This is where the drive team members come in. While the main driver focuses on the team's robot, the drive team members can warn of possible scoring from opponents.

For the last seconds of the game, we have assumed that a double park will currently be improbable. Resultantly, only one robot in the alliance will be parking. If our alliance is capable of doing so, we will then take on the task of attempting to score more blocks, or preventing the opposing side from scoring blocks themselves.

Brainstorming

September 8, 2025

Written by: A.B., J.C.

Present: J.C., T.C., A.B.

Goal

We want our robot to be able to move fast around the field in order to obtain blocks quickly because having possession of blocks will allow us to score and there is no limit to how many blocks we can possess. We need a mechanism to intake the blocks into our robot and transport them to another system which could be made up of chain or rubber bands that will act as an elevator or rope to transport the blocks up the robot and be shot out from the top into the goals. In addition to the scoring mechanism, we need another sub system that allows us to expand outwards to reach and remove the blocks from the vertical block loading stations.

Deciding Systems

Our robot will need a drivetrain to traverse the field, an intake to take possession of the blocks, an elevator or chain mechanism to move the blocks up the robot, and some sort of expansion that will allow our robot to reach into the loaders.

Method for Decision Choice

- In order to make choose between possible ideas, we have decided to incorporate a decision matrix.
- Each member will get to rate ideas (1-5) based on different criterion factors such as time to build, reliability, and resource cost.

Example

	Criterion Point 1	Criterion Point 2	Criterion Point 3	Criterion Point 4	Total
Idea 1	4,4,5,3,4	2,2,3,2,3	1,2,3,2,2	4,3,3,4,4	60
Idea 2	1,3,2,2,2	1,1,1,2,4	2,3,3,2,3	5,4,4,5,4	54
Idea 3	3,4,4,4,2	5,3,3,4,5	4,2,2,3,3	5,3,4,4,3	70
Idea 4	5,4,5,3,3	3,3,3,4,3	4,4,3,4,3	5,4,3,4,3	73

Brainstorming: Structure

September 8, 2025

Written by: J.C.

Present: J.C., T.C., A.B.

Subsystem 1: Drivetrain

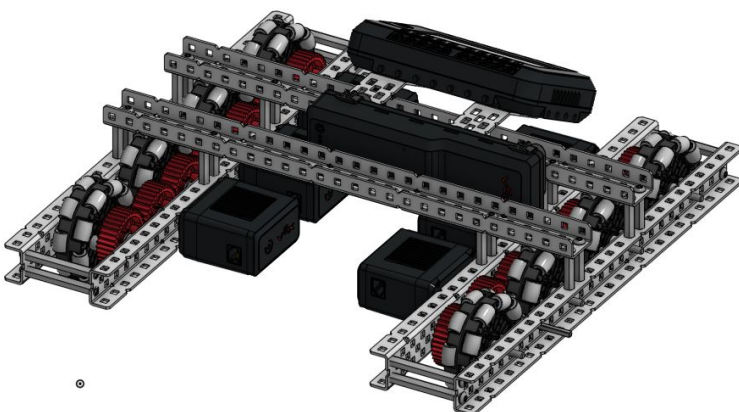
Idea 1 - Eight Wheels

For our first drivetrain design idea, we could use eight 2.75" omni-wheels and 6 blue motors (that have a 36:1 gear ratio) all connected by fourteen 32T High Strength gears for a 1:1 gear ratio on a 13.5"x13.5" drivetrain. This drivetrain design would make our robot extremely fast, which would allow us to traverse the field quickly and reach blocks before other teams do. However, we currently don't have enough 2.75" omni-wheels to complete this drivetrain so we would need to order parts.

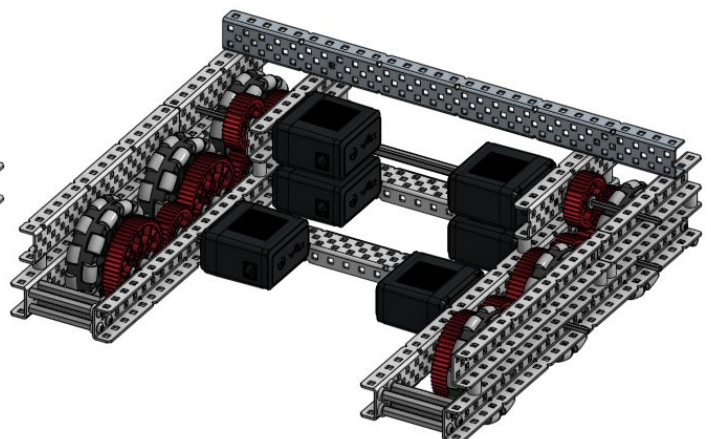
Idea 2 - Six Wheels

Another idea we have, is using 6 3.25" omni-wheels and 6 blue motors (that have a 36:1 gear ratio) where the back 2 on both sides of a 13.5"x14.0" drivetrain are stacked to free up space in the front of the robot for an intake and elevator. These wheels and motors will be connected by 12 gears, six 60T High Strength gear for each wheel and six 32T High Strength gears for each motor. The resulting drivetrain would have a 32:60 gear ratio, which is geared for torque, allowing our robot to hold its ground and prevent being easily pushed – a crucial advantage when scoring and defending against other robots. Although the drivetrain would be geared for torque, the blue 36:1 motors would counterbalance this torque ratio by increasing the speed at which the torque gear ratio spins, allowing us to be both tenacious and nimble.

Idea 1



Idea 2



Brainstorming: Structure

September 8, 2025

Written by: J.C.

Present: J.C., T.C., A.B.

Deciding on the Drivetrain

- As a reminder, each member rating the ideas based on an agreed upon criteria
- Rating is done on a scale of 1-5
- The idea with the most points at the end wins

	Easy to Build	Resource Efficient	High Speed	High Torque	Structural Integrity	Total
Eight Wheel Drive	3,2,2,3,2	3,2,3,3,3	5,5,4,5,4	1,1,2,2,2	4,4,3,4,5	77
Six Wheel Drive	4,3,4,3,3	1,1,1,2,4	3,3,3,4,3	4,3,3,3,4	5,4,4,3,5	80

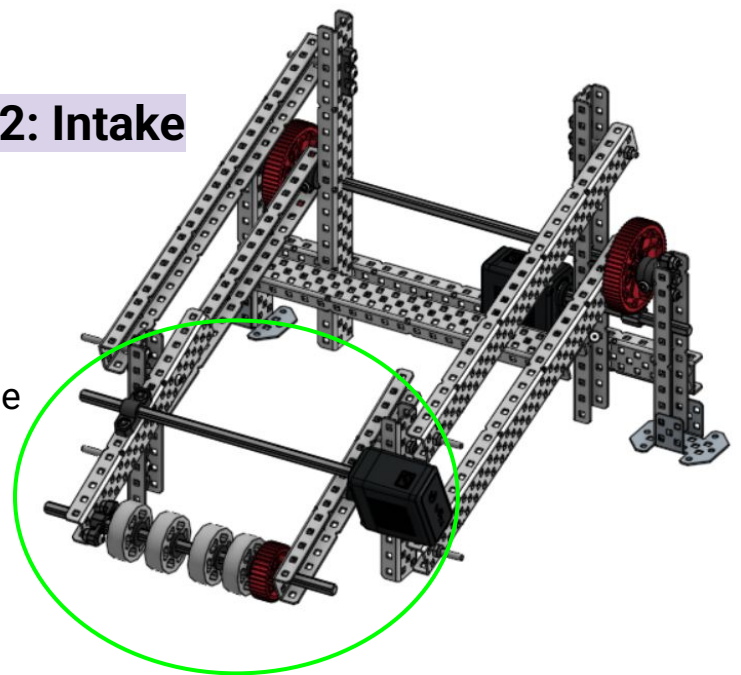
Selecting Drivetrain Idea

While the ideas were very close in rating, we don't have enough resources to test both out physically. Therefore, we are going directly with the idea that won: The six wheel, speed driven, torque geared drivetrain.

Subsystem 2: Intake

Idea 1 - Flex Wheels

We could use flex wheels on a high-strength shaft. This will function like a roller, and using the tangential force in the counterclockwise direction (flex wheels rolling inward), allow our robot to intake the blocks and push them into an elevator or chain system to move up the robot.



Brainstorming: Structure

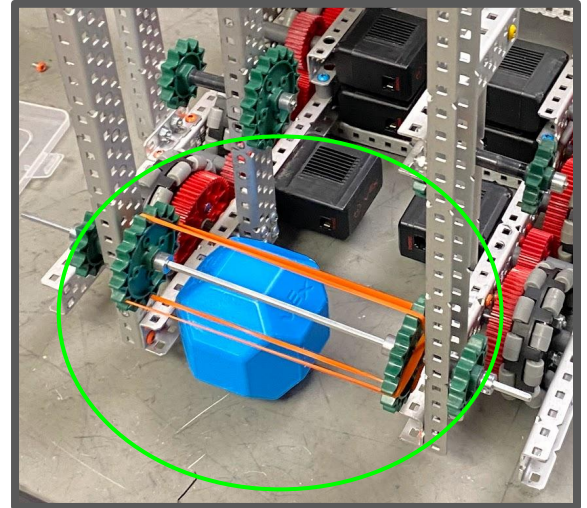
September 8, 2025

Written by: A.B., J.C.

Present: J.C., T.C., A.B.

Idea 2 - Rubber Bands

We can create a vertical elevator with an intake directly attached at the bottom. This intake will be comprised of sprockets inserted on a horizontal shaft. The sprockets will be held in place by shaft locks and will have rubber bands stretching across them, creating a more flexible tangential force.



Deciding on the Intake

	Easy to Build	Good Block Grip	Space Conserving	High Speed	Total
FlexWheels	4,4,3,4,3	4,3,3,4,4	2,1,2,1,1	4,4,3,4,4	62
Rubber Bands	5,4,5,5,4	3,3,3,4,3	5,5,4,5,4	3,4,4,3,3	79

Selecting Intake Idea

Here, the flex wheel system was initially rated high in many categories. However, it was considered to take up too much space, which was the deal breaker that gave the win to the rubber band intake.

A.B. 9/8/25

Brainstorming: Structure

September 8, 2025

Written by: J.C.
Present: J.C., T.C.

Subsystem 3: Transport

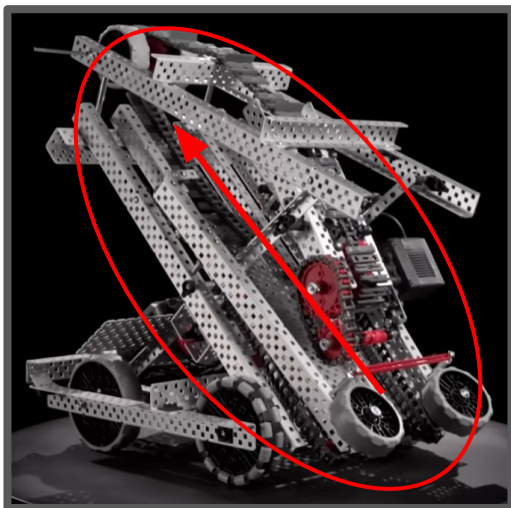
Idea 1 - Ramp

The first idea we have for transporting the blocks upwards is utilizing a ramp with chain that will pull the blocks upwards and shoot them through the top of the ramp. This idea stems from the Vex Push Back Hero Bot Dex. This ramp mechanism will allow us to score by transporting the blocks upwards to reach the goal height, however, this mechanism takes up a lot of space and does not allow us to store many blocks.

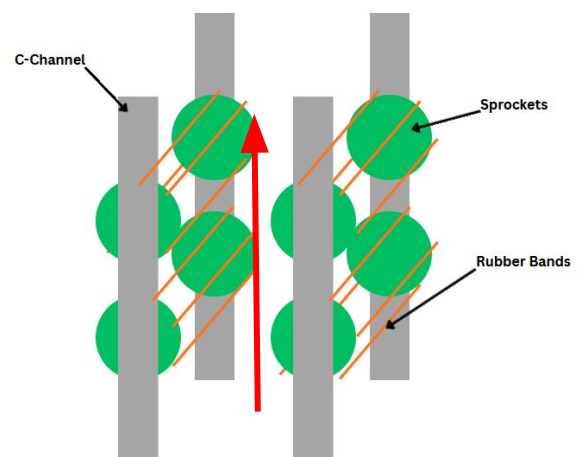
Idea 2 - Elevator

We will create an elevator mechanism that utilizes functions similar to our intake. There will be two pairs of parallel vertical c channels, with each pair containing 2-3 rows of sprockets. Each row of sprockets will be placed on a regular horizontal shaft and will have multiple rubber bands stretching across them. These sprockets are intended to function much like rollers, and will therefore require to have high amounts of rubber bands in order to increase contact time with the blocks. All together, this will allow us to have a conveyor/elevator mechanism that can efficiently move multiple blocks along a vertical axis.

Idea 1



Idea 2



Brainstorming: Structure

September 8, 2025

Written by: A.B., J.C.

Present: J.C., T.C., A.B.

Deciding on the Transport

	Easy to Build	Block Mobility	Space Conserving	Block Space	Total
Ramp	3,3,2,2,2	5,4,4,4,4	2,1,1,2,1	3,2,2,3,4	54
Elevator	5,5,5,4,4	3,4,4,3,3	4,4,3,2,3	3,4,4,3,4	74

Selecting Transport Idea

We decided constructing idea 2 for the Elevator mechanism because it requires less space on the robot which will leave space in the back for a basket to store blocks. In Push Back there is no limit to how many blocks you can possess, therefore having high possession of both your blocks and the opponents blocks is crucial.

Subsystem 4: Match Loader

What is it?

The match loader is a pneumatics system which allows for the intake of blocks from the loader. This system is very feasible, as models are usually compact enough to fit within the 22 inch extended robot size limit, and it requires very simple coding and materials such as ABS, C channels, and shafts. The benefits of a system like this are mainly in its ability to intake from the loaders. The loaders are preloaded with 6 blocks, 3 of the alliance's color. This is very beneficial when it comes to situations such as during the autonomous period, when having more blocks available is an asset. Cons of the match loader, however, go back to its size. While true that the system *is* compact enough to fit within nearly any robot, it does add more volume, which increases difficulty in movement, as well as the chances for immobilization.

Brainstorming: Structure

September 8, 2025

Written by: A.B.

Present: J.C., T.C., A.B.

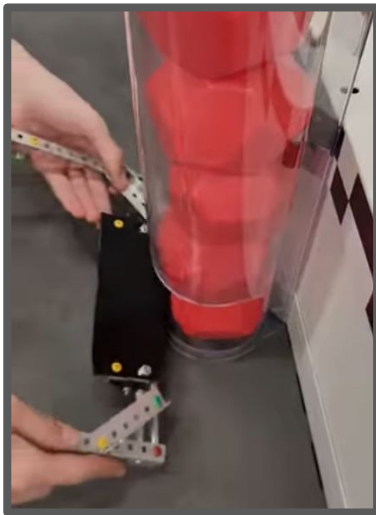
Idea 1 - ABS

Use ABS plastic as the material for the center part of the match loader. The main point of using ABS here, is to allow a smoother intake of the blocks.

Idea 2 - Standoffs

Use stand offs as the material for the center part of the match loader. This is usually best when fabrication time or resources are scarce, as it doesn't require the extensive measuring and cutting that ABS needs.

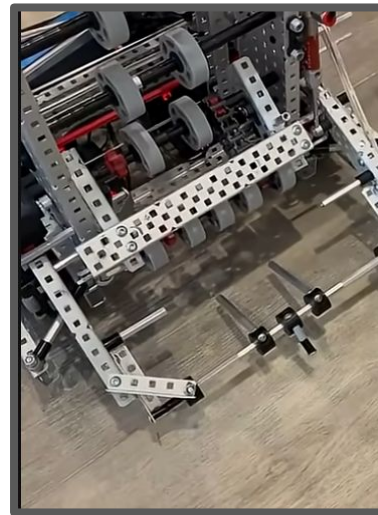
Idea 1



https://www.youtube.com/shorts/be_liFO82SY

From @21510R-GraveyardShift

Idea 2



<https://www.youtube.com/watch?v=0emHKpEN2bw>

From @76058A

Deciding on the Match Loader

	Consistent	Easy to Build	Easy to Align	Total
ABS	5,4,4,4,5	2,3,2,3,3	5,5,5,4,5	59
Standoffs	4,3,3,4,4	4,4,3,3,4	2,3,2,2,2	47

Brainstorming: Structure

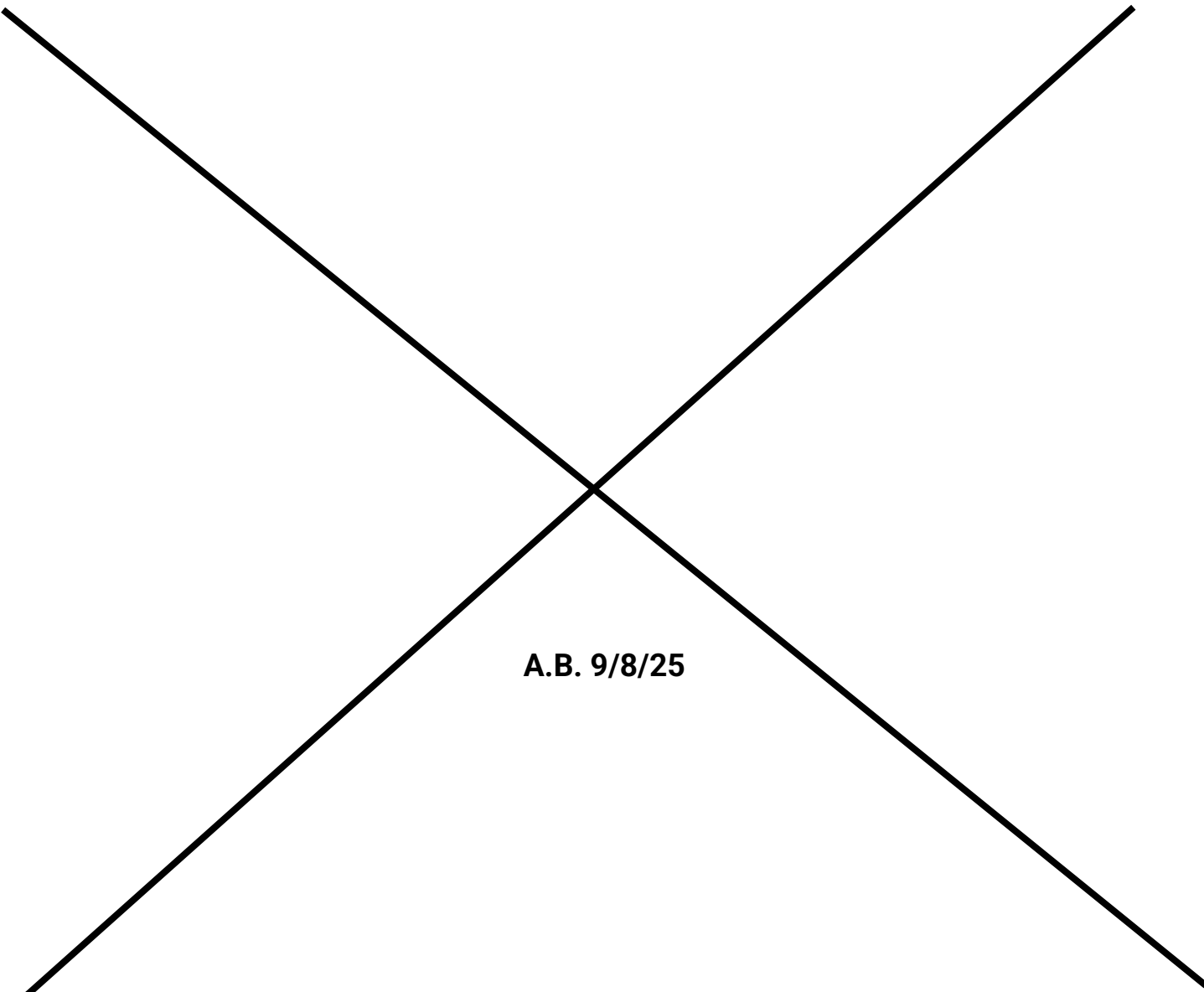
September 8, 2025

Written by: A.B.

Present: J.C., T.C., A.B.

Selecting Match Loader Idea

Since our team has sufficient resources, and is aiming to optimize the loader intake, going for the more flexible and forgiving ABS intake material may be the best option. Though it may take a bit more time to fabricate, it should ultimately prove to have been worthwhile, by allowing swifter and smoother block loading than stand offs ever could.



A.B. 9/8/25

Design Conclusions

September 8, 2025

Written by: A.B.

Present: J.C., T.C., A.B.

Drivetrain

Utilizes six omni wheels, a low to high gear ratio, and blue motors. Has great Torque and the necessary speed to keep it balanced.

Intake

Uses rubber bands stretching across two sprockets in order to grip and pull blocks into the robot. However, durability has not yet been fully tested.

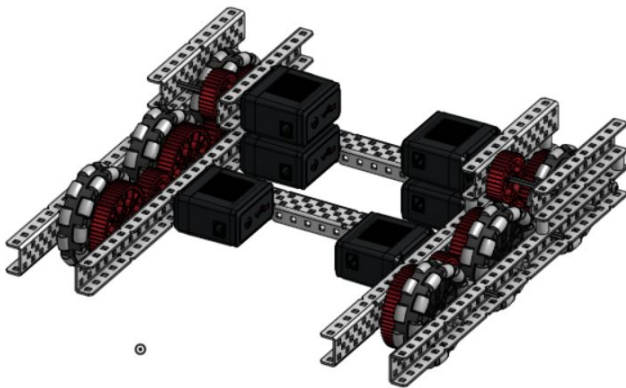
Transport

Composed of a vertical elevator with rows of sprockets that connect with rubber bands. This saves on space, allowing us to create a dedicated storage system.

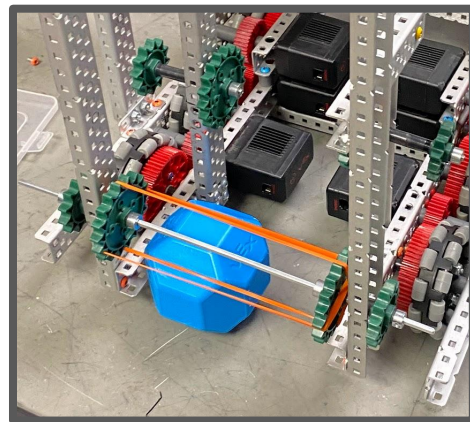
Match Loader

Made up of an ABS centerpiece for smoother, more consistent, match loading.

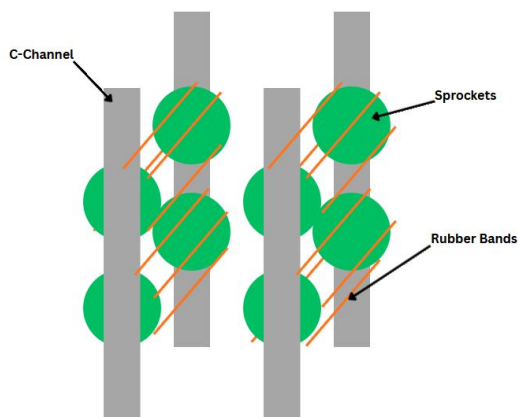
Drivetrain



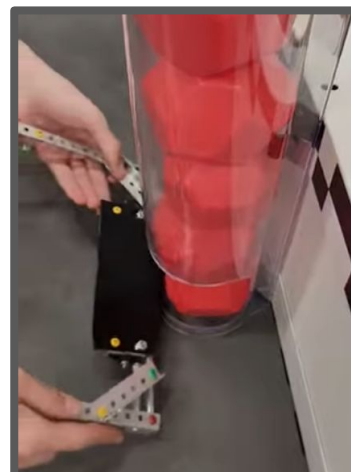
Intake



Transport



Match Loader



Building the Drivetrain

September 16, 2025

Written by: D.M.

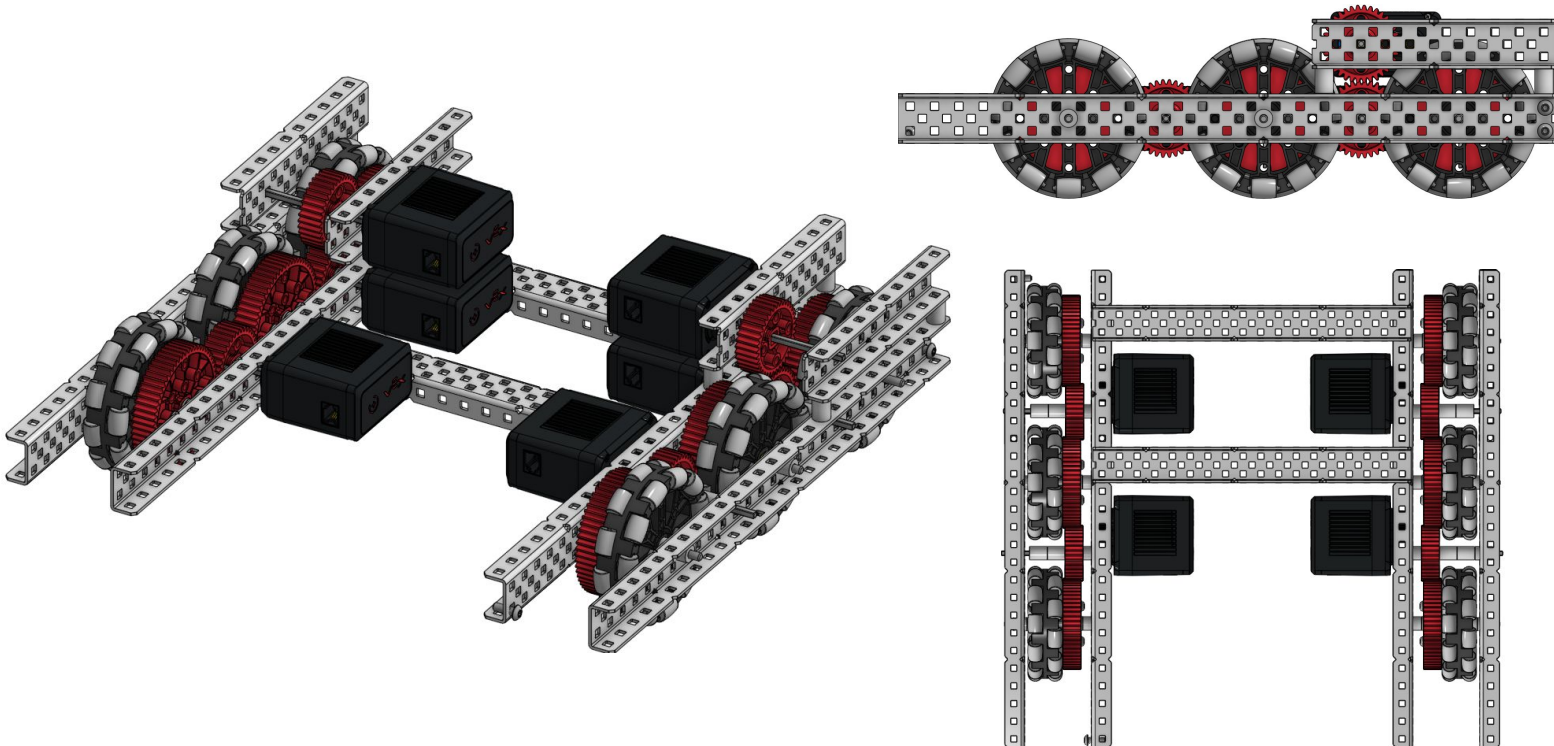
Present: J.C., T.C., D.M., A.B., D.S.

Goal

- *Build a wheel system that allows the robot to properly move in all directions once connected with coding to a robot controller.*
- *Format the wheel system properly, in order to more easily achieve competition tasks such as parking.*

Design

The drivetrain consists of two identical wheel systems that are separated by an eight inch gap (Equivalent to a 17 Hole C-Channel). Individually, each wheel system is composed of two 13.5 inch C channels (32 hole C channels) that protrude outward and are connected by four 2 inch standoffs in the front. There are three evenly spaced omnigears held together by a series of three size 60t gears and connected through a series of three 32t gears. Each omni wheel is directly connected to their gear.



Improving the Drivetrain

September 20, 2025

Written by: D.M., A.B.

Present: J.C., T.C., D.M., A.B., D.S.

Goal

- *Improve drivetrain stability*

Reason for Modification

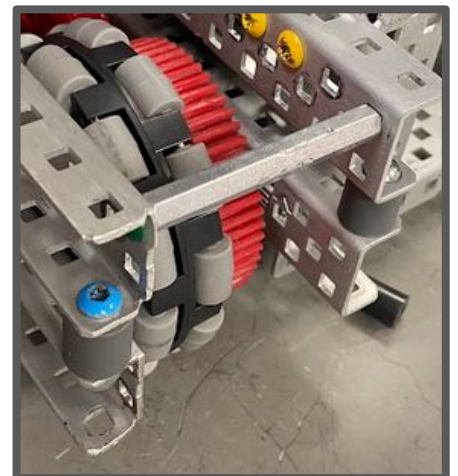
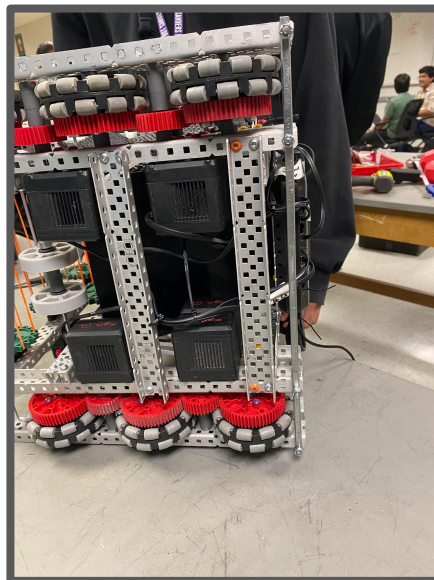
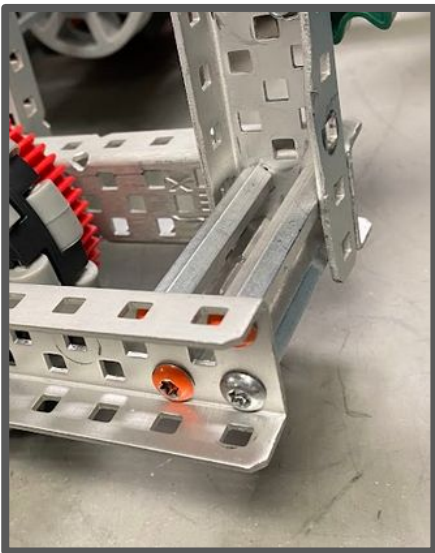
Our drivetrain suffers from inward bending. This is because the center of the robot is too heavy and the weight is not properly distributed throughout the entire drivetrain. This not only hinders our ability to drive accurately, but also our ability to meaningfully expand upon our robot.

Notes

- We are not able to run a full c channel across the entire drivetrain
- We are trying to add components, rather than removing them from our drive

Solution

To fix the drivetrain bending issue, we have decided to first individually stabilize both sides of the drivetrain. Each side is composed of two parallel c channels, which we can connect using standoffs (2 in) and spacers. Afterwards, we can secure the whole structure using a high strength shaft that runs *beneath* the drivetrain frame. The high strength shaft will act like a rigid structural brace, reducing the degrees of freedom of the drivetrain components.



Building the Intake & Elevator

September 23, 2025

Written by: D.M., T.C.

Present: J.C., T.C., D.M., A.B., D.S.

Goal

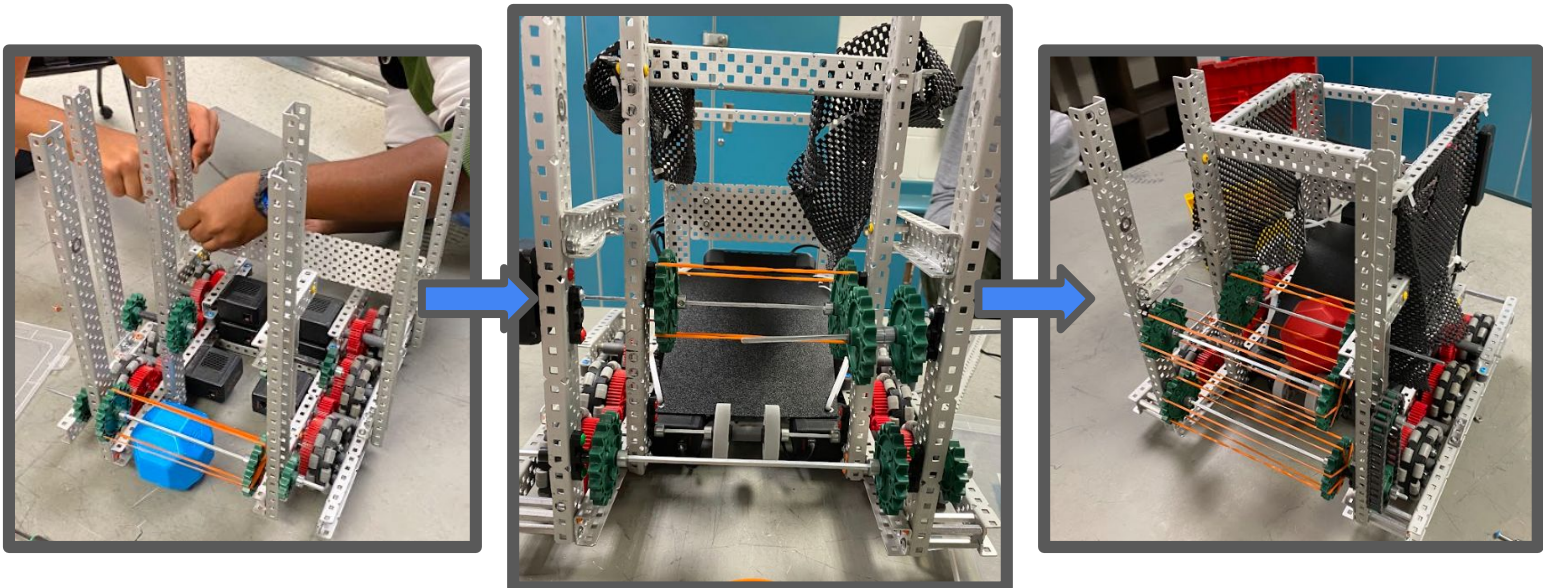
- *Use the chosen designs as blueprints, and build our intake and transport mechanisms into the robot*

Purpose & Design Reasoning

- The intake mechanism is strategically part of the elevator mechanism, which saves up on space for our storage system
- The final system should be able to do the following things:
 - Intake blocks
 - Store blocks
 - Outtake the stored blocks
 - Move blocks on a vertical axis
 - Dispense blocks at the top of the mechanism

Materials

- (4) 15 inch aluminum c channels
- (6) 6 inch shafts
- Sprockets
- Rubber bands



Building the Intake & Elevator (Cont.)

September 23, 2025

Written by: D.M., T.C.

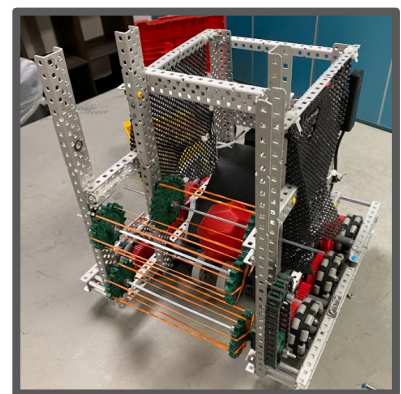
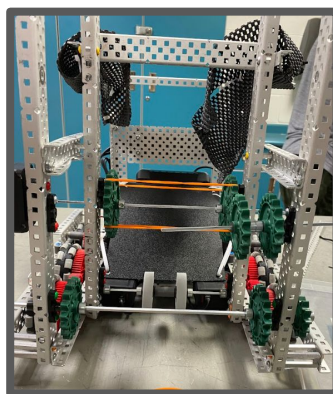
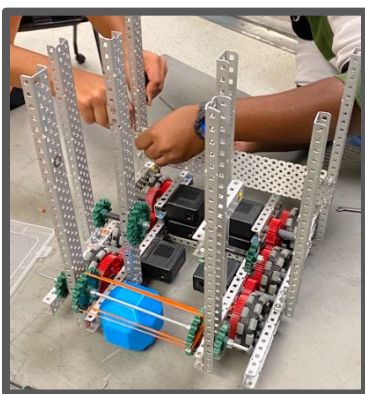
Present: J.C., T.C., D.M., A.B., D.S.

Goal

- *Identify and implement a solution to prevent the rubber bands on the intake system from tearing or snapping during operation.*

Design Change

We noticed that the thin orange rubber bands were ripping after about two days of field use, so we switched to the thicker orange bands, which lasted much longer and improved intake reliability. We also rearranged the gears and added chains to keep the intake system synchronized. After that, we installed flex wheels to help guide the blocks, but the first design caused the wheels to block the cubes from entering the basket. To fix this, we added a motor to the flex-wheel shaft, allowing the wheels to actively pull the blocks smoothly into the basket. Finally, we redesigned the top section of the intake. Our team tested several ideas and made small adjustments until we found a version that consistently fed blocks into the goal efficiently.



Building the Cage

September 29, 2025

Written by: D.S.

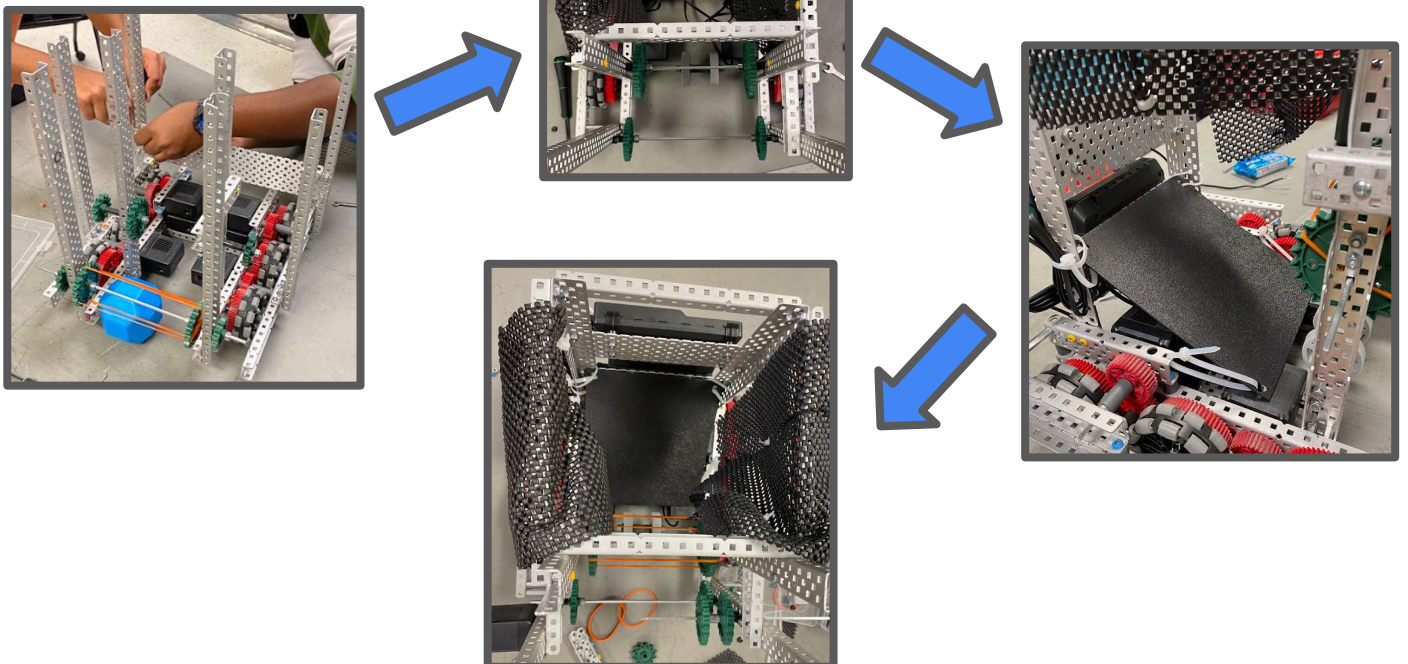
Present: J.C., T.C., D.M., A.B., D.S.

Goals

- *Build a compartment (cage) to securely hold blocks during intake and scoring.*

Design Change

To create a functional intake system, we needed a compartment capable of holding blocks without letting them fall out. We began by mounting two 2×30 C-channels at the eighth hole of the drivetrain C-channels, then added 2×25 long angles two holes from the end to form the cage structure. Next, we attached mesh along the sides to prevent blocks from slipping out during movement. To complete the design, we installed a slanted ABS plastic panel at the back of the intake. This angled surface helped guide blocks smoothly into the cage and made out taking more efficient by allowing the blocks to slide out cleanly when the intake reversed.



Designing the Dispenser

October 1, 2025

Written by: D.M.

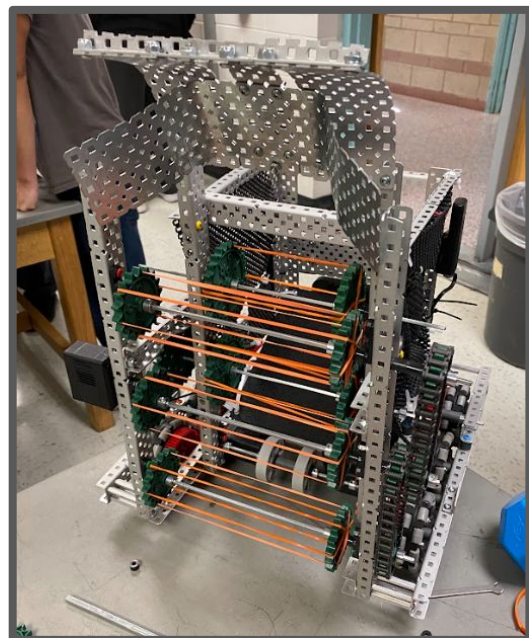
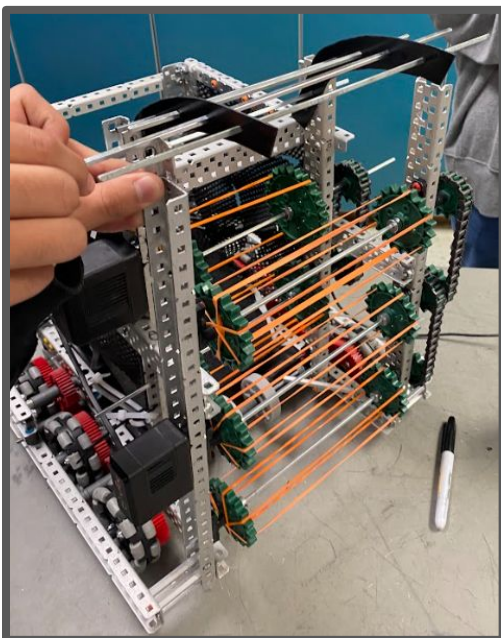
Present: J.C., T.C., D.M., A.B., D.S.

Goals

- *Design a curved guide piece made from ABS or aluminum to improve the outtake process for the blocks.*

Design Change

We made several adjustments to the intake system, including changes to the gears and rubber bands mentioned earlier. As shown in the images on this page, our team redesigned the top section of the intake by adding gears, rubber bands, and a shaft. However, the shaft was free-spinning, which caused the mechanism to stop functioning correctly. To fix this issue, we connected the top gears to a chain system linked to another gear already powered by a motor in the intake. This allowed the entire top assembly to spin consistently, improving both intake and outtake performance.



Dispenser Prototyping

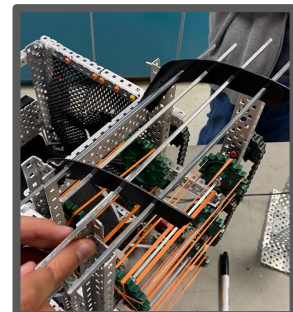
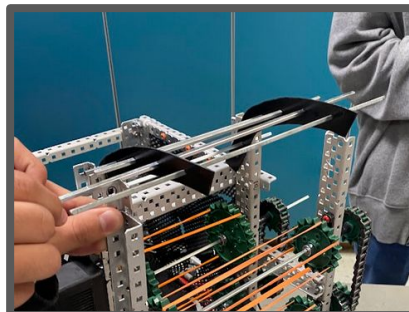
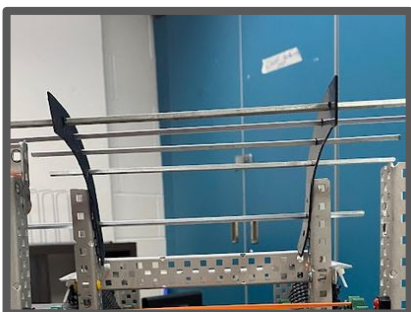
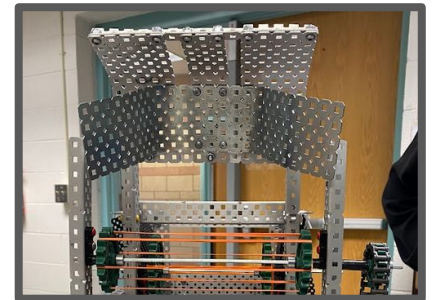
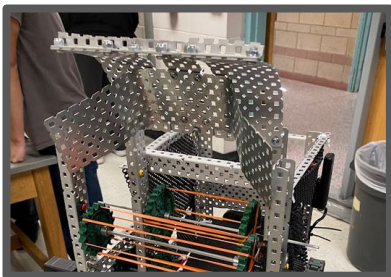
October 1, 2025

Written by: D.M.

Present: J.C., T.C., D.M., A.B., D.S.

Design Change

While designing the hood (a bent aluminum plate) for the top of the intake, we created several prototypes to improve how blocks were guided into the goal. Our first prototype used a bent aluminum hood with side plates. However, this design caused blocks to shoot out toward the sides, and they often jammed inside the hood, which limited our scoring consistency. For the second prototype, we replaced the aluminum hood with a curved ABS cutout and drilled a metal drive shaft through the ABS to help guide the blocks. This version performed better, but it still struggled to push the blocks fully into the goal. This led to our final working prototype, where we added a single bent aluminum plate to help push blocks deeper into the goal. Although this improved performance, we wanted an active mechanism to assist with outtaking. We built a system using sprockets and rubber bands, which successfully pushed blocks out but had one major issue—the mechanism free-spun. To finalize the design, we connected a chain and sprocket to the shaft holding the mechanism. This allowed the outtake system to spin with the motor instead of free-spinning, completing our final, reliable design.



Dispenser Prototyping (Cont.)

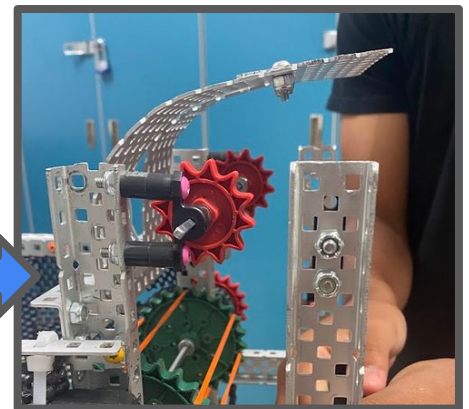
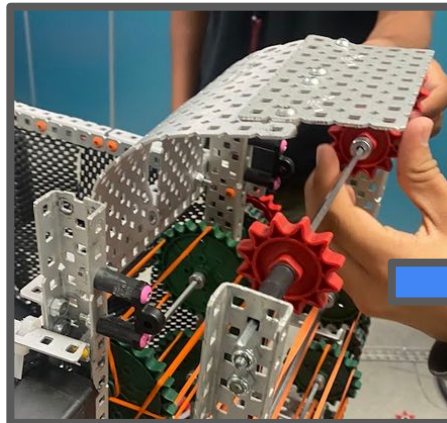
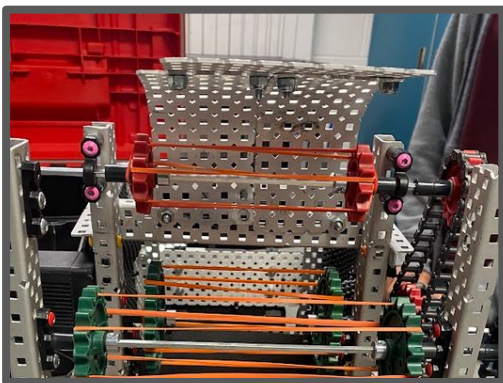
October 4, 2025

Written by: D.M.

Present: J.C., T.C., D.M., A.B., D.S.

Design Change

When designing the mechanisms that would help spit out the blocks we had two goals in mind the first one being of course the main objective that it would help spit out the blocks, and the second one being that it should help push the blocks further in the goal, with the objective to make more space to fit and score more blocks in the goal. As I previously mentioned our team first just made the mechanism using just a metal drive shaft, two sprockets and rubber bands. When we attached it to the robot it would just free spin. To fix the issue we decided to attach a sprocket to the metal drive shaft, as for the sprocket that was attached we decided to connect it to the chain system that was already made for the intake. Everything was going smoothly until we noticed that the blocks would not spit out properly, to fix this issue we pushed the whole system towards the the hood, moving it from the front of the hood to where the hood was exactly attached (View the image as it show how we switched the system). This solution helped our team complete the two goals which was; it would help spit out the blocks more smoothly and efficiently, and the second being help spit out the blocks with more force to make space for more blocks to be scored by us.



Improving the Cage

October 28, 2025

Written by: T.C.

Present: T.C.

Goals

- *Reshape the ABS to better control the flow of balls entering and exiting the robot.*

Design Change

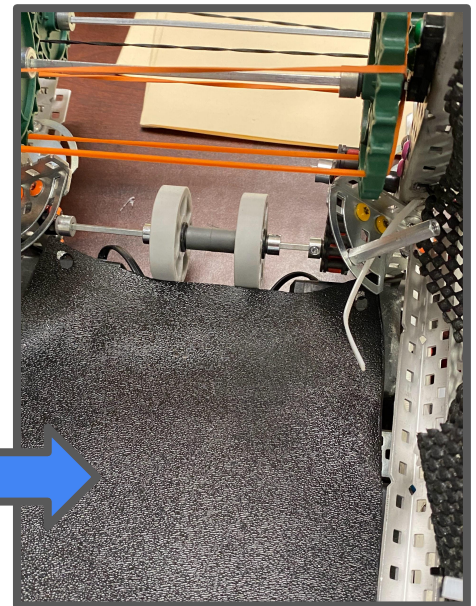
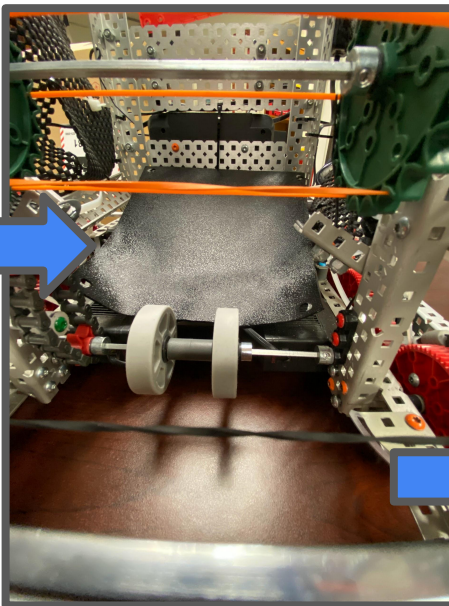
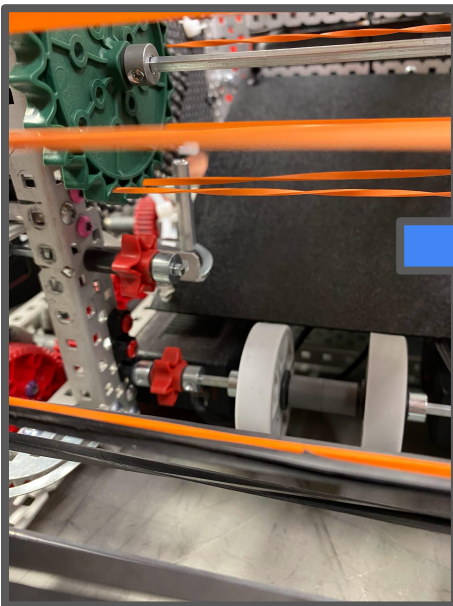
We heated the ABS using a heat gun set to 300 °C, which softened the material enough for us to mold it into a more curved, funnel-like shape. This reshaping allowed us to create a smoother path for the balls during both intake and outtake.

Benefits

The improved curvature helps the balls flow more easily without getting stuck, especially during autonomous. This increases the reliability of both our autonomous routine and the robot's overall performance.

Before

After



Improving the intake

October 28, 2025

Written by: D.M.

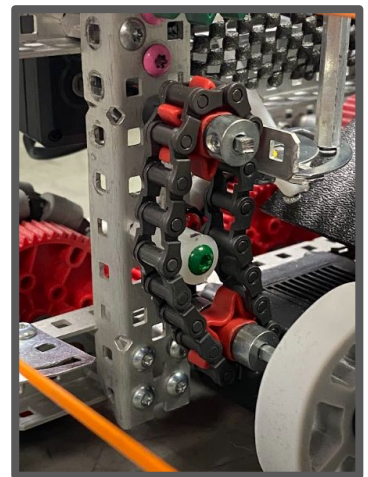
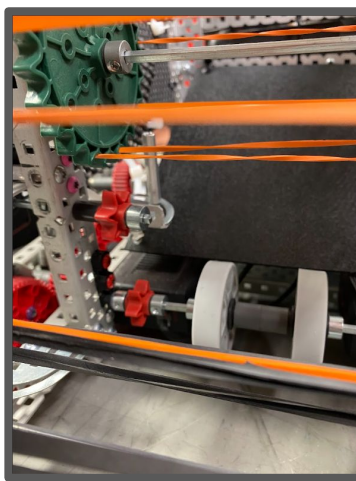
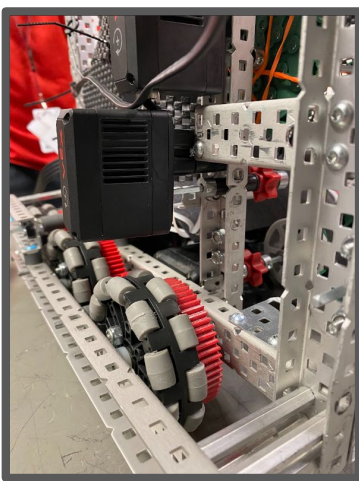
Present: J.C., T.C., D.M., A.B., D.S.

Goals

- *Tweak the intake so blocks can move smoothly between the cage and the elevator without getting stuck.*

Design Change

To improve block movement, we expanded our use of flex wheels. At first, we installed the flex wheels directly, but they free-spun, which prevented the blocks from entering the basket. We attempted to fix this by adding a motor; however, removing the shaft during installation was difficult, though the setup eventually worked. Next, we connected the motor to the flex wheels using sprockets and a chain so the wheels would actively spin. The problem was that the chain was either too loose or wouldn't connect properly, causing inconsistent motion. To solve this, we added a spacer on the C-channel near the motor. The spacer pressed lightly against the chain, pushing it outward just enough to simulate proper tension. This eliminated the slack and allowed the chain to run smoothly and consistently.



Trenton Tournament: VEX V5 Push Back Blended Tournament

Nov.1 - Notes and Reflection

November 1, 2025

Written by: D.M., J.C.

Present: J.C., T.C., D.M., A.B., D.S.

Results

In this competition our team analyzed our robots performance, looked for areas with flaws, and most importantly looked over the matches along with what other robots did. In summary for this competition we won a total of five matches out of eleven. Three of the matches we won were for the qualifiers, which we didn't qualify however our sister team (597A) alliance with our team. Both sister team and our team won R16 and quarter final, during semi-final it is were we lost.

Observations

Competition

- During most of our matches our team noticed that some robots had a Descoring Mechanism.
- The Descoring Mechanism had change the field during the game since it would take out all the blocks that had be scored in the top goal.
- Lastly we noticed that most people had a chained ramped.

Our Team

Successes

- The robots elevator/intake system worked flawlessly.
- The robot was able to play strong defense. Our team was able to make it to the Semi-Finals

Failures

- Our Robots code was not always accurate and would sometimes score and sometimes it wouldn't.
- Our power cord was very loose and disconnected from our battery during the semifinals.
- When the robot tipped over near the top goal it would get stuck.

Rank	28
WP / AP / SP	6 / 35 / 131

<

597C - Robobusters ...

☆

@ Trenton Tournament: VEX V5 Push Bac...

Q4

9:12 AM

4148B

597A

56

11

2148S

597C

Q11

9:40 AM

12169A

597C

35

0

48546P

2148C

Q22

10:24 AM

1585Z

597D

69

19

597C

3350Z

Q32

11:04 AM

597E

597C

30

67

2821C

597F

Q45

11:56 AM

48546A

3350A

12

66

597C

597B

Q52

12:54 PM

43272Y

4148A

28

22

23835B

597C

Q61

1:30 PM

597C

2148G

3

134

9189X

1585Y

Q67

1:54 PM

23835A

43272X

53

64

1585X

597C

R16 4-1

4148B

1585X

57

59

597A

597C

QF 2-1

43272Y

43272U

20

75

597A

597C

SF 1-1

9189X

1585Z

124

14

597A

597C

Qualification Ranking

Current Rank

WP / AP / SP / W-L-T

28

6 / 35 / 131

3-5-0

Skills

Skills Rank

Score / Attempts

20

4 / 1

<> 0 / 0

Modifying the Outtake

November 5, 2025

Written by: A.B.

Present: A.B. J.C. D.S. D.M. T.C.

Goals

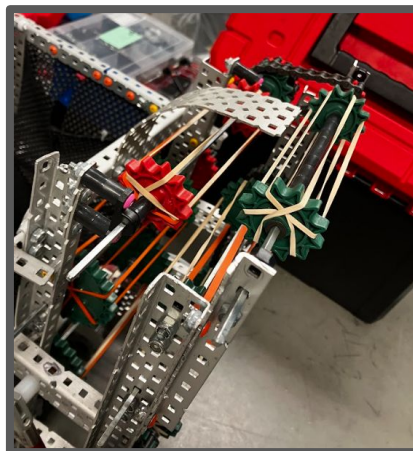
- *Modify the outtake system so the robot can dispense balls with greater force and require less precise alignment.*

Reasoning

During our November 1st competition, we experienced several issues that limited our scoring potential. Although our outtake functioned moderately well, it did not generate enough power to push blocks deep into the goals. As a result, we often placed multiple blocks inside the goals but failed to secure control bonuses, which ultimately cost us several matches. Another major issue was alignment. Our current outtake requires very precise positioning before scoring, which is difficult to achieve during a match with three other robots moving around the field. This extra alignment time reduced our scoring efficiency and consistency.

Redesign

To address these problems, we are adding an additional set of sprockets with rubber bands to the outtake system. This upgrade increases the kinetic energy applied to each block, allowing the robot to outtake with more force and reducing the need for perfect alignment.



Phoenix Premiere Robotics Blended Tournament: Early Season Push Back Nov.8 - Notes and Reflections

November 8, 2025

Written by: J.C.
Present: J.C., T.C., D.M., A.B., D.S.

Strategy Evaluation

Our game strategy during this competition was to score on the outer (top) goal and lower center goal because our elevator mechanism scoring mechanism was effective at scoring in those. However, we lacked the ability to score high center goal, so we focused on the other two and played heavy defense when our alliance had an effective scoring robot. We made sure to park at the end of each round. This worked half the time as seen in our qualifier win to loss ratio, where we went 3-3.

What Worked? & What Didn't Work?

Our elevator mechanism was very effective at scoring both top (outer) goal and lower center goal. Although we were able to score effectively, our robot lacked a match load and high center goal scoring so we lost out on the opportunity to score high center and collect additional blocks from match load tubes. A lot of teams also utilized a descoring mechanism which greatly affected the outcome of our matches.

Changes to Make

Although our basket design was effective at storing blocks, it took a while before we were able to score them so we will choose to redesign our robot in the future. The most important changes we need to make right now are adding a descoring mechanism and a match load mechanism to our robot as they are very powerful in the push back meta.

Phoenix Premiere Robotics Blended Tournament: Early Season Push Back							
Qualifier #9	Nov 8th at 10:21 AM	8188N	597C	39	77711J	3028G	11
Qualifier #18	Nov 8th at 11:28 AM	10066A	8188K	87	597C	3691C	40
Qualifier #22	Nov 8th at 11:50 AM	597C	106Z	102	8188S	2148F	3
Qualifier #36	Nov 8th at 1:24 PM	20480H	4148B	124	597C	8188M	0
Qualifier #47	Nov 8th at 2:02 PM	597C	77711F	8	9189X	3691A	132
Qualifier #55	Nov 8th at 2:32 PM	597C	4148S	38	597A	597B	24
R16 #3-1	Nov 8th at 3:32 PM	28006C	106Z	107	597C	597B	38
Rank	20						
WP / AP / SP	6 / 15 / 86						

Pneumatics Part 1 - Brainstorming

November 10, 2025

Written by: D.M.

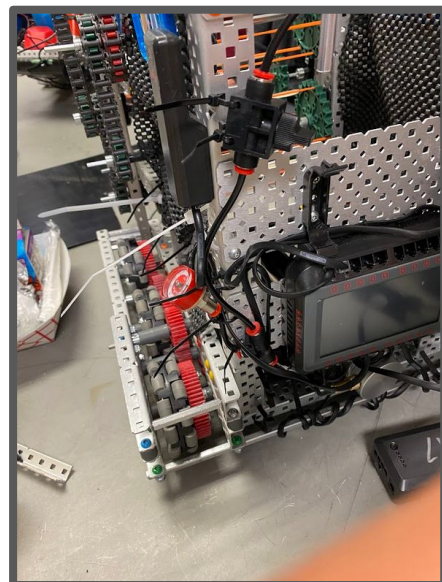
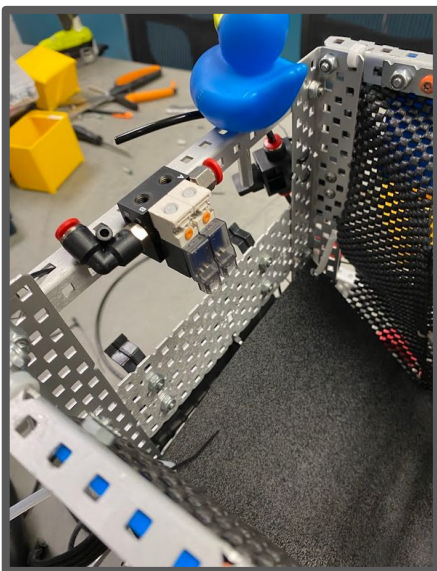
Present: J.C., T.C., D.M., A.B., D.S.

Goals

- *Add a pneumatics-powered descoring mechanism to remove opponents' blocks from the goal, increasing our robot's defensive capabilities.*

Design Change

After our competition at Trenton High School, we identified a major weakness in our defensive strategy. During one of our matches, we successfully scored five blocks into the tall goal. However, our opponent used a unique side-mounted descoring mechanism that quickly knocked all of our blocks out of the goal. This single mechanism completely shifted the match and demonstrated how vulnerable we were to defensive counter-play. In response, our team decided to integrate a pneumatics system to power our own descoring mechanism. Pneumatics provide fast, controlled, and high-force movement, making them ideal for pushing opponent blocks out of the goal before they can secure control bonuses. Today, our team captain began assembling the pneumatics system that will serve as the foundation for this new defensive tool.



Pneumatics Part 2 - Adding Tanks

November 11, 2025

Written by: A.B.

Present: A.B., T.C., D.M., D.S.

Goals

- *Modify our air-tank system to provide enough stored air for the pneumatic mechanisms to operate reliably throughout a match.*

Design Change

Our robot originally used a single small air tank for the pneumatic system. During testing, we found that this setup did not supply enough air to power the mechanism consistently, especially during repeated activations. While a larger tank would solve the issue, it could not fit within the robot's available space. To increase air capacity without increasing the footprint, we added a second small tank connected in series with the first. This doubled our usable air supply while maintaining a compact layout that fits cleanly inside the robot.

Benefits

With the expanded air capacity, our pneumatic mechanisms can operate reliably throughout the entire match without the risk of running out of air. This upgrade significantly improves the consistency and effectiveness of our descoring system.



Pneumatics Part 3 - Descoring & Match Loader Mechanism

November 12, 2025

Written by: D.M.

Present: J.C., T.C., D.M., A.B., D.S.

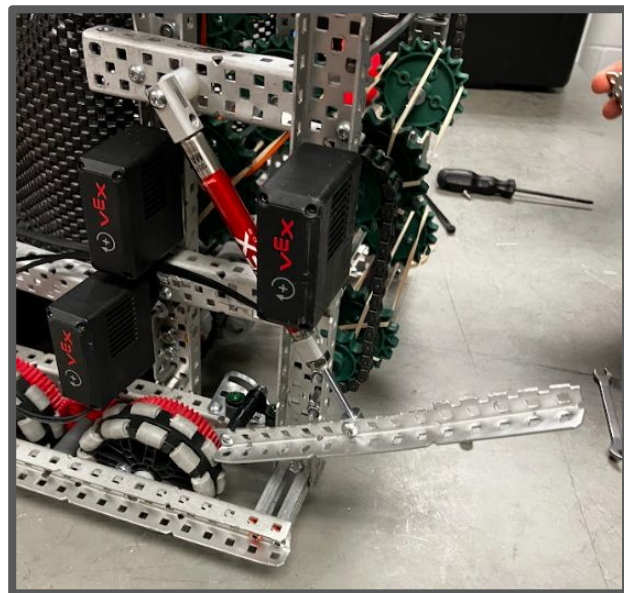
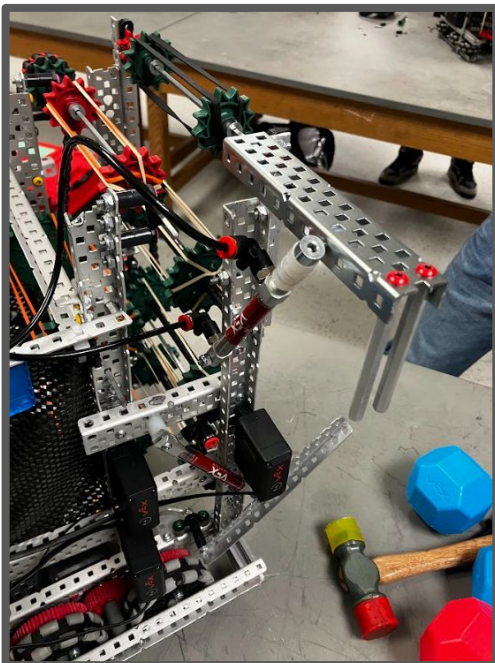
Goals

- *Complete and refine the descoring mechanism, ensuring it functions reliably before competition.*

Design Change

Today we fully attached both the descoring mechanism and the match-loading mechanism to the robot. For the descorer, we mounted a cut-out C-channel onto the pneumatic piston. The piston connects to the tubing system, which is controlled by a solenoid. The solenoid regulates airflow, allowing precise activation of the descoring mechanism during matches. We also integrated the match-loading mechanism into the same pneumatic system. This mechanism allows us to unload blocks from the field's match-loading tubs more efficiently. Now that we can reliably reach and release blocks from the tubs, our scoring potential increases significantly

Descoring Mechanism



Match Loader Mechanism

SSHS Robotics Rumble [Blended]: Push Back Edition Nov. 15 - Notes and Reflection

Nov 15, 2025

Written by: D.M., J.C.

Present: J.C., T.C., D.M., A.B., D.S.

Results

SSHS Robotics Rumble [Blended]: Push Back Edition

Qualifier #8	Nov 15th at 9:50 AM	597F	106X	70	597B	597C	37
Qualifier #14	Nov 15th at 10:13 AM	20550X	597C	13	10066A	2148D	100
Qualifier #23	Nov 15th at 10:44 AM	1585Y	46790B	31	597C	90219A	39
Qualifier #33	Nov 15th at 11:21 AM	597C	46790A	49	90219C	2148F	13
Qualifier #37	Nov 15th at 11:36 AM	597C	4148A	11	1585X	106A	88
Qualifier #45	Nov 15th at 12:01 PM	597A	2148B	29	4148S	597C	63
Qualifier #52	Nov 15th at 1:05 PM	597C	597E	50	2750H	1585Z	73
Qualifier #60	Nov 15th at 1:31 PM	13974A	2750B	63	4148B	597C	43
R16 #7-1	Nov 15th at 2:57 PM	2821C	4148S	72	597C	1585Y	45
Rank	22						
WP / AP / SP	6 / 30 / 227						

In this competition we encountered difficulties with our radio as we kept disconnecting during our matches. We thought this was most likely an issue with the competition system rather than our radio, however after our matches we noticed that our radio cable was loosely connected.

Observations

Competition

- Our team noticed that our radio from the robot disconnected then reconnected.
- Most teams used a conveyor belt with chains and flaps.

Our Team

Successes

- Our modified outtake on the top helped us push more blocks on the top goals

Failures

- Our teams robots chain fell off
- Robots radio was malfunctioning
- The robot would still get stuck near the top goal when tipped.

Harvest Bots - Miller Grove Tournament: Miller Grove HS Tournament Nov. 21 - Notes and Reflection

Nov 21, 2025

Written by: D.M., J.C.

Present: J.C., T.C., D.M., A.B., D.S.

Results

Harvest Bots - Miller Grove Tournament: Miller Grove HS Tournament

Qualifier #4	Nov 21st at 9:44 AM	30119A	75149A	19	32410H	597C	28
Qualifier #13	Nov 21st at 10:23 AM	32410A	597C	9	597F	42329A	49
Qualifier #29	Nov 21st at 11:22 AM	75117B	20480H	30	2750A	597C	11
Qualifier #39	Nov 21st at 12:10 PM	75503P	62714B	48	75426B	597C	0
Qualifier #44	Nov 21st at 1:32 PM	597C	42329C	0	89278A	30119B	38
Qualifier #53	Nov 21st at 2:07 PM	75503X	9950X	36	32410E	597C	13
R16 #5-1	Nov 21st at 3:33 PM	89278A	75160X	33	62714B	597C	3

Rank 39

WP / AP / SP 2 / 10 / 39

This was our worst competition to date. We had major issues with the robot brain after another robot smashed itself onto our robot's brain, leaving our robot's brain unable to function properly.

Observations

Competition

- There were mostly basket robots and some ruiguan robots.

Our Team

Our team was unable to perform our usual abilities of scoring, parking, and defense because our robot brain was damaged and our robot could not function properly and our controller lost connection constantly.

Successes

- Reached the Team Selection and Round of 16.

Failures

- Our damaged brain costed us the competition. It made it difficult to perform well as our robot could not function properly.

Vex V5 Push Back Redesign Research

December 1, 2025

Written by: D.M.

Present: A.B., T.C., D.M., D.S.

Goals

Our goals for today is to research robot designs and use a decision matrix to help us choose a solid design. Meanwhile the team just made sure the robot was ready for our next competition.

Design Change

There was no change for the design or the robot. Today our team was researching robot designs through videos and CADs we found. The top two designs that our team we're most interested in, were the S shape robot and the Ruiguan designs. We eventually decided to focus on researching multiple Ruiguan designs. When researching we put either the source link or video link along with the type of robot and a description of its abilities.

Vex V5 Push Back Design Research								
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C4	A	B	C	D	E	F	G	H
1	Link	Designs	Pros	Cons	CAD?	Materials		
2	https://www.youtube.com/watch?v=2J2D_8QCcs	Ruiguan	Fast at scoring and moving, the liners allow for easy scoring	Can score mid, but requires pistons	Time to find a cad, or make one if we're unlucky			
3	https://www.youtube.com/watch?v=pmtiZW-W120	Ruiguan (American)	Basically everything above					
4	https://www.instructables.com/3D-CAD-Design-ed-VEX-V5-Robot-for-Competitive-Robot/	Similar to ruigans intake						
5								
6	https://www.youtube.com/watch?v=k6ENT2IOWJY&t=288s	S shape				We should have everything, but there a small omni wheel at the bottom	They use six 11 wat motors for the drivetrain and one 11 and at least one 5.5 wat motor on the intake	
7								
8								
9	https://www.youtube.com/watch?v=2J2D_8QCcs							
10								
11								
12								
13								

Video Analysis

December 1, 2025

Written by: D.M.

Present: A.B., T.C., D.M., D.S.

Goals

Our goals for today is to research robot Ruiguan Design through a video of a person going into detail on its design.

Quick Summary on video

The video talks about the Ruiguan robot. The interviewer explains that their robot is very fast, lines up quickly, and scores really well compared to other teams. Harrison explains how their drivetrain, intake, and shooter were designed to focus on fast cycling instead of holding a lot of balls. He also talks about using flaps, polycarbonate pieces, and an aligner to make scoring more accurate and consistent. The team made changes based on testing and competition experience to improve speed and reliability. Overall, the video shows how smart design and strategy helped the team use the robot perform extremely well.



RuiGuan Subsystems Breakdown

December 1, 2025

Written by: J.C.

Present: A.B., T.C., D.M., D.S.

Goals

- *Our goals for today is to research robot Ruiguan Design through a video of a person going into detail on its design.*

Subsystem 1: Drivetrain

The drivetrain subsystem is responsible for moving the robot around the field so it can collect blocks and position itself to score. It typically consists of multiple VEX motors connected through gears to axles that drive the wheels. The chassis frame supports these components and keeps them aligned. Most Ruiguan designs use a four-motor or six-motor tank drive with traction or omni wheels to balance speed, control, and maneuverability during matches.

Subsystem 2: Flexible Intake

The intake subsystem collects blocks from the field and pulls them into the robot. It is usually located at the front of the robot and uses spinning rollers made from flex wheels or rubber bands powered by motors. Side guides or funnels help direct blocks toward the rollers so they enter the robot smoothly. The intake is designed to be wide and fast so the robot can gather multiple blocks quickly while driving. The loose chain allows for the intake to bob up and down as it intakes blocks.

Subsystem 3: Elevator Ramp

The block transfer or conveyor subsystem moves blocks from the intake to the scoring mechanism through the interior of the robot. It typically consists of belts, chains, or rollers powered by motors that continuously move blocks along a straight path from the front to the back or top of the robot. Structural guides keep the blocks aligned so they do not jam. This system allows blocks to flow efficiently through the robot, enabling fast scoring cycles.

Subsystem 4: Outtake Flap/Soring Mechanism

The scoring or outtake subsystem delivers blocks into the goal. It is usually located at the end of the conveyor system and uses rollers or belts powered by motors to push blocks out of the robot. A hood or angled plate often guides the blocks so they enter the goal accurately. This subsystem ensures that blocks are released smoothly and consistently during scoring.

Subsystem 5: Match Loader

The match loader system helps the robot quickly receive blocks from the human loader during the match. It usually includes a wide intake area with funnels or guide plates that direct the blocks into the intake rollers. Once inside, the conveyor moves the blocks through the robot toward storage or scoring. It is comprised of intake rollers, guiding structures, and the conveyor system.

Subsystem 6: Descore Mechanism

We will utilize our previous pneumatics descoring mechanism design on this new robot.

Start of robot rebuild

December 3, 2025

Written by: A.B. and D.M

Present: J.C., T.C., D.M., A.B., D.S.

Goals

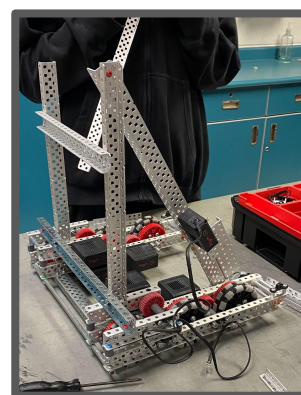
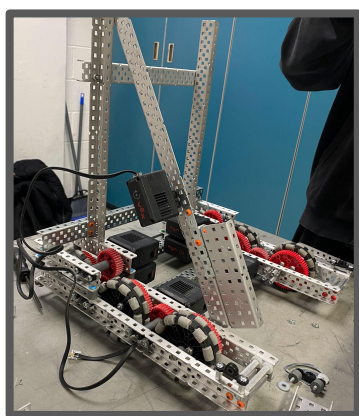
- *Create a new robot framework inspired by recent high-performing designs, allowing us to shift our overall game strategy.*

Notes

- We chose to keep our original drivetrain because it already performs at a high level. Its gearing provides strong torque, and the use of blue motors gives us the speed needed to balance that torque. Since the drivetrain is both reliable and efficient, we decided to focus our redesign efforts on the robot's upper structure

Design Change

We are currently developing a new frame shaped like a ramp, inspired by robots such as the HeroBot Dex and Ryugan-style designs. These robots use an escalator-based intake and scoring system that moves blocks smoothly from one end of the robot to the other. While the concept is simple, our priority is precision and durability. It is not enough for the frame to work once it must remain stable and consistent throughout the entire season. Our focus is on building a strong, well-aligned structure that supports the escalator mechanism, maintains rigidity under load, and ensures smooth block movement. This new framework will allow us to adopt a more efficient scoring strategy and significantly improve our competitive performance.



Flexible Intake Subsystem

January 2, 2026

Written by: D.M., J.C.

Present: J.C., T.C., D.M., A.B., D.S.

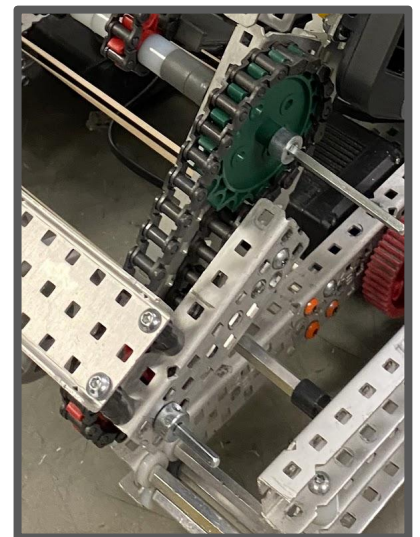
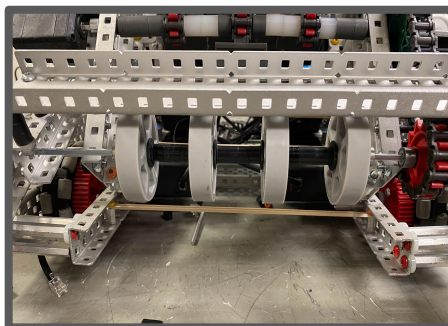
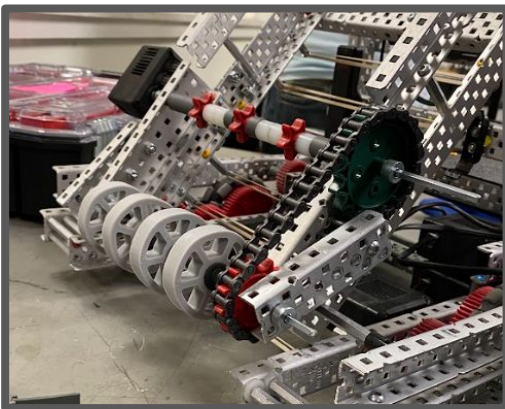
Goals

- *Modify the intake to increase its range of motion and prevent blocks from getting stuck during the intake process.*

Design Change

To improve reliability, the build team added flex wheels separated by spacers on a long metal shaft. This shaft is driven by a small gear that is chained to a larger gear, which connects to the shaft powering our conveyor belt. This setup ensures that the intake and conveyor remain synchronized.

The key improvement in this design is its flexibility. The large sprocket connected to the smaller sprocket has intentional slack in the chain, allowing the intake to pivot up and down as it collects blocks. This added range of motion prevents blocks from jamming and helps the intake adapt to different angles during gameplay



Ramp Elevator Subsystem

January 2, 2026

Written by: D.M.

Present: J.C., T.C., D.M., A.B., D.S.

Goals

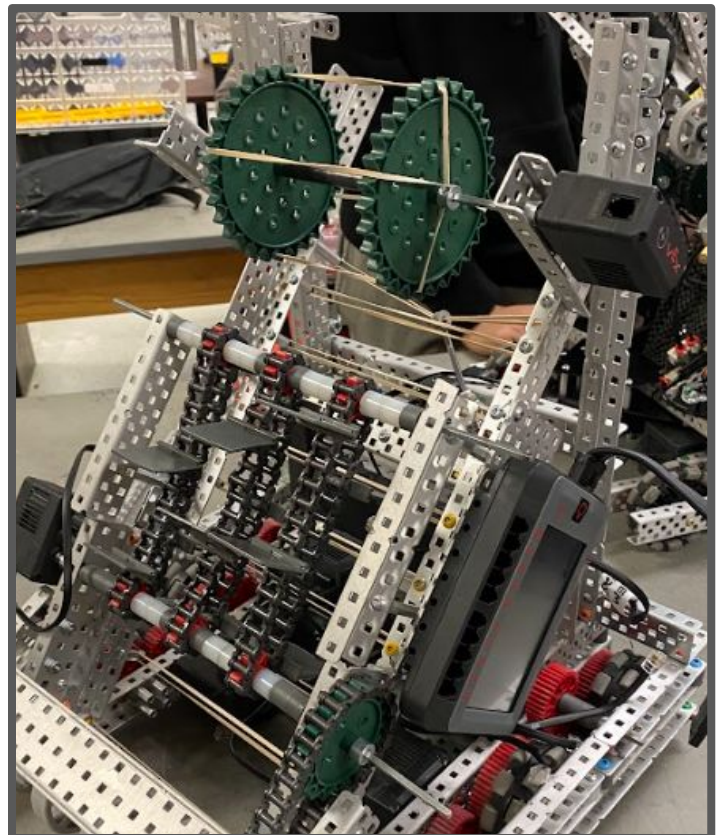
- *Our goals for today's work day was to Install a ramp-style structure that lifts blocks upward toward the future outtake mechanism.*

Design Change

To support the new ramp system, our team attached a small C-channel to each side of the structure, creating secure mounting points for two shafts. Each shaft held three evenly spaced small sprockets, separated with precise spacers to ensure smooth rotation and alignment. One shaft was positioned near the top of the ramp and the other near the bottom, close to the intake, forming the foundation of the elevator-style mechanism.

We also installed a chain system with attached flaps to move the blocks upward in a controlled, elevator-style motion. As the chain rotates, the flaps scoop the blocks and guide them smoothly up the ramp toward the future outtake.

The entire system is powered by a green motor mounted at the bottom of the ramp. This motor drives the lower shaft and its sprockets, which in turn power the chain. With this setup, the robot can reliably lift blocks upward without jamming or losing alignment. These upgrades transformed the ramp into a consistent, elevator-style mechanism that prepares the robot for a high-performance outtake system.



Outtake/Flap Subsystem

January 5, 2026

Written by: D.M.

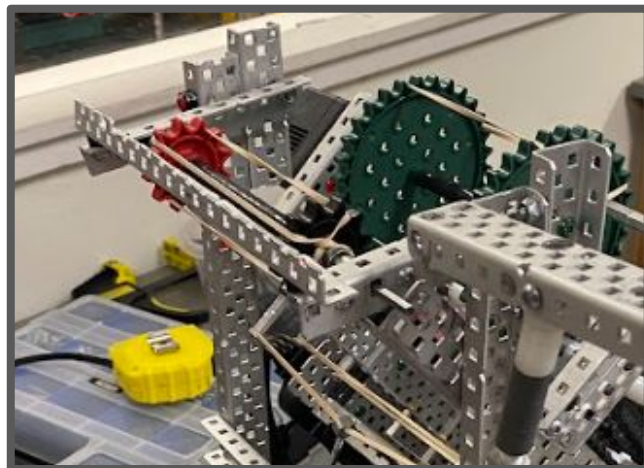
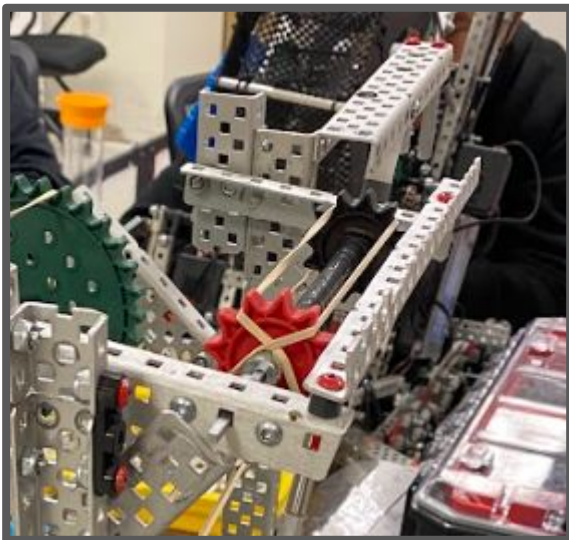
Present: J.C., T.C., D.M., A.B., D.S.

Goals

- *Our goals for today were to Finish and refine the outtake flap to increase scoring speed and accuracy during matches.*

Design Change

To improve the robot's outtake performance, we added a sprocket-driven rubber-band mechanism that provides a controlled and consistent release of blocks. The system is mounted on a long shaft and secured with two shaft locks to ensure stability during high-speed operation. This upgraded design allows the outtake flap to push blocks out smoothly while maintaining strong, repeatable force. As a result, the robot can score faster and with greater precision, significantly improving our overall scoring efficiency and reliability during competition



Matchload Subsystem

January 5, 2026

Written by: A.B. and D.M

Present: J.C., D.M., A.B., T.C., D.S.

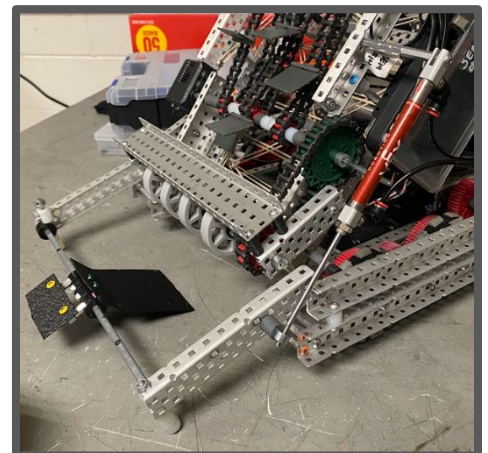
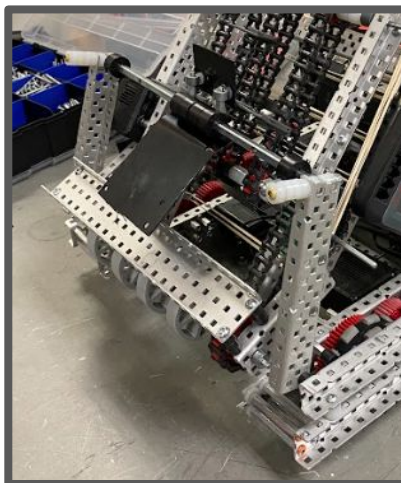
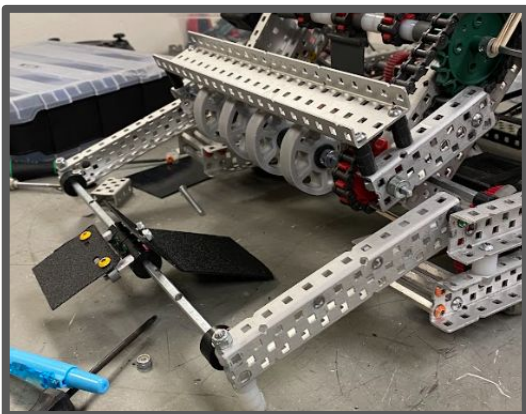
Goals

- *Begin developing a match-load system that allows the robot to efficiently extract blocks from the field loaders.*

Reason for Implementation

Throughout the first half of the season, we observed that many of the highest-scoring robots consistently used match-load systems. These mechanisms gave teams a major advantage by allowing them to quickly acquire blocks during both driver control and autonomous. Robots equipped with match-load systems were able to gather large numbers of blocks in a short amount of time, significantly increasing their scoring potential. Recognizing this competitive advantage, we decided to begin designing our own match-load system to elevate our robot's performance and keep pace with top-tier teams.

Potential Ideas



Trenton Winter Classic HS VEX V5

Pushback Jan.10 - Notes and Reflection

January 10, 2026

Written by: D.M., J.C.

Present: J.C., T.C., D.M., A.B., D.S.

Results

Our team made it to the round of 16 where we ultimately lost. However, we did earn the Design Award.

Observations

Competition

- *Most of our opponents move fast*
- *During this competition our aptitudes and flaws are the opposite of what they were originally. At the beginning of the year, our basket bot succeeded due to its consistent outtake and block maneuverability. Now, our rebuild Ruigan robot succeeds more with its intake, match loading, and descroing mechanism, while lacking consistency with the outtake.*

Our Team

Our robots had both ups and downs for example during a match A block had gotten stuck, which lead our captain to create a funnel out of rubber bands.

Successes

- Added Autonomous code to our robot
- Made the intake more smooth with the modification of a funnel.
- We fixed the conveyor belt by adding spacer, which gave more space for the blocks to not get stuck.

Failures

- Our robot got stuck on a block and blocks kept getting stuck in the intake
- The funnel wasn't always working until we modified it again.
-

< 597C - Robobusters ... ☆					
@ Trenton Winter Classic HS VEX V5 Pus...					
Matches					
Q3	43272X	42	8	597C	
9:08 AM	597E			41411C	
Q19	597D	23	24	43272E	
10:12 AM	597C			78987A	
Q26	597C	30	14	67679B	
10:40 AM	70291A			20480C	
Q39	20480K	48	66	43272U	
11:32 AM	597C			12169Z	
Q47	597F	5	36	20480H	
1:04 PM	30119B			597C	
Q59	43272S	30	28	41411B	
1:52 PM	75503Y			597C	
R16 7-1	43272S	33	54	597F	
	597C			70291G	
Qualification Ranking					
Current Rank			WP / AP / SP / W-L-T		
29			4 / 20 / 126		
			2-4-0		

Outtake Flap Redesign

January 13, 2026

Written by: J.C. and D.M

Present: J.C., A.B., T.C., D.S.

Current Design

Our current outtake flap uses two 6T sprockets powered by a blue motor through a chain-driven 12T-to-6T gear ratio, optimized for speed.

Issues to Address

We identified two major limitations with the existing design:

- **Poor contact with blocks:** *The sprocket shaft sits too high on the outtake flap, making it difficult for blocks coming up the elevator ramp to engage with the rubber-banded sprockets. This reduces the force and consistency of each outtake.*
- **Missing secondary sprocket set:** *Many high-performing teams use two pairs of sprockets to increase contact time and push blocks deeper into the goal tube. Due to limited sprocket availability, we cannot replicate this setup directly.*
- **Lack of an ABS pushing surface:** *Other teams successfully use an ABS panel at the end of their outtake flap to guide blocks farther into the goal tube. This has proven highly effective, and our current design lacks this feature.*

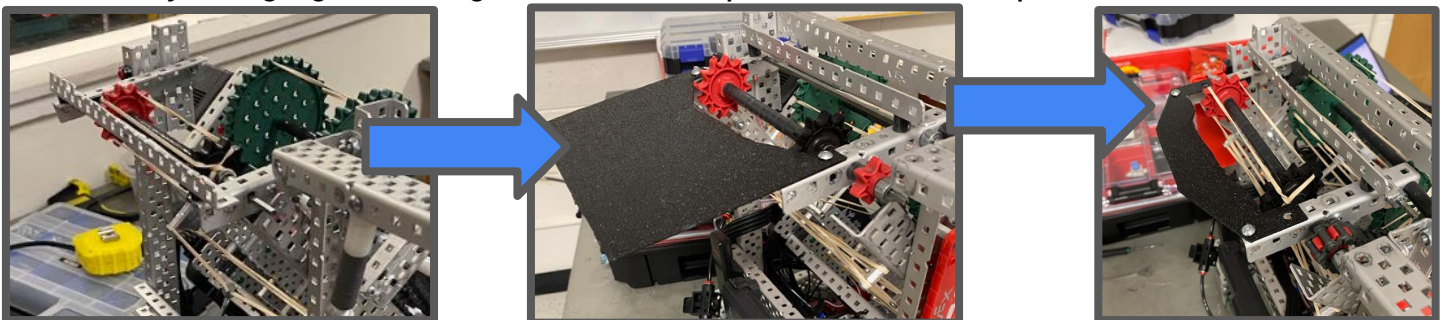
Changes

To address these issues, we are implementing two key modifications:

1. Lowering the sprocket shaft: *We will shift the sprocket shaft down by two holes on the outtake flap. This brings the sprockets closer to the top of the elevator ramp, ensuring more reliable contact with the blocks as they exit the ramp.*

2. Adding a trapezoid-shaped ABS flap: *We will construct a trapezoid ABS panel and mount it to the end of the outtake flap. This ABS extension will slide into the goal tube and help push blocks farther inside, increasing scoring depth and consistency.*

These changes will significantly improve block engagement, outtake force, and scoring reliability, bringing our design closer to the performance of top-tier robots.



Aligner Redesign

January 13, 2026

Written by: D.M.

Present: J.C., A.B., T.C., D.S.

Previous Design

Our earlier aligner design used a triangular ABS piece mounted to the C-channel with two short screws. A second identical ABS triangle was attached using standoffs, creating a rigid, single-piece alignment structure.

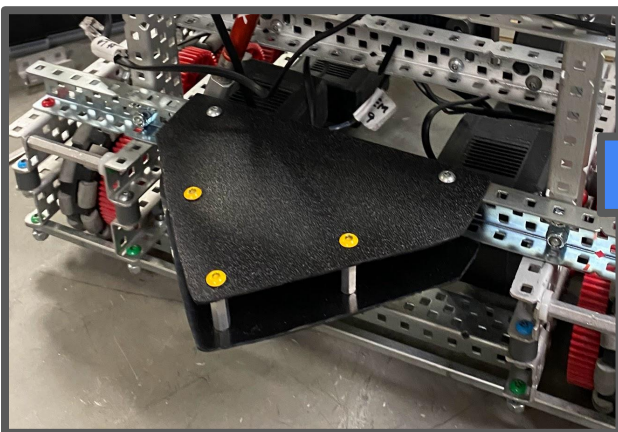
Issues to Address

This design presented two major problems:

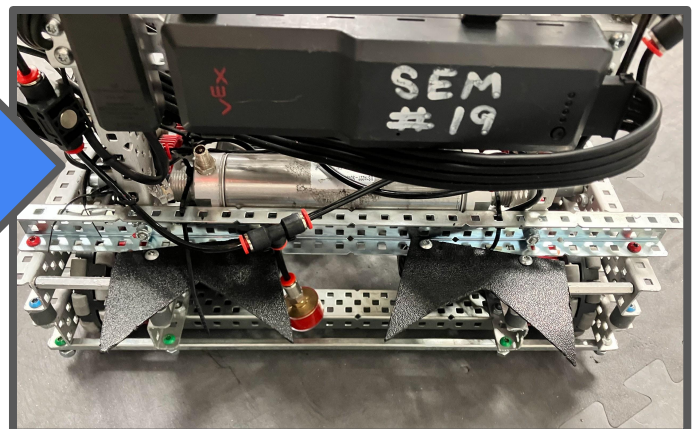
- **Exceeded ABS size limits:** *The combined ABS structure extended beyond legal material constraints, making it non-compliant.*
- **Inconsistent alignment:** *The aligner did not reliably latch onto the top goal, causing frequent misalignment during scoring. This inconsistency slowed down our drivers and reduced our scoring efficiency. These issues directly impacted match performance and needed to be resolved.*

Changes(current Design)

To address these problems, we redesigned the aligner as a two-part system. This new configuration is smaller, fully within legal ABS limits, and easier to mount. Each piece was cut with a precise triangular indentation that hooks smoothly onto the top-goal legs. This allows the robot to align quickly and stay locked in place even when opponents apply defensive pressure. The new design provides consistent, reliable alignment and significantly improves our ability to score efficiently under match conditions.



OLD DESIGN



NEW DESIGN

Ramp Funnel

January 13, 2026

Written by: D.M.

Present: J.C., A.B., T.C., D.S.

Current Design

Our current ramp-funnel design uses rubber bands arranged in an X-shaped web to keep blocks centered as they travel upward. These rubber bands are mounted to standoffs and secured with small screws

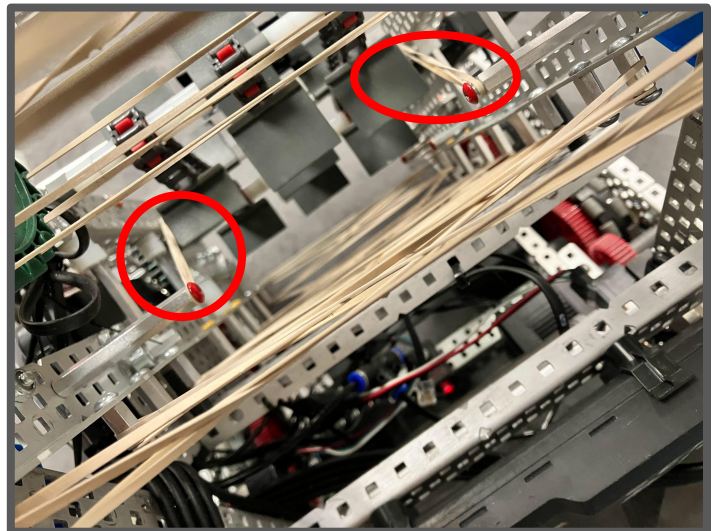
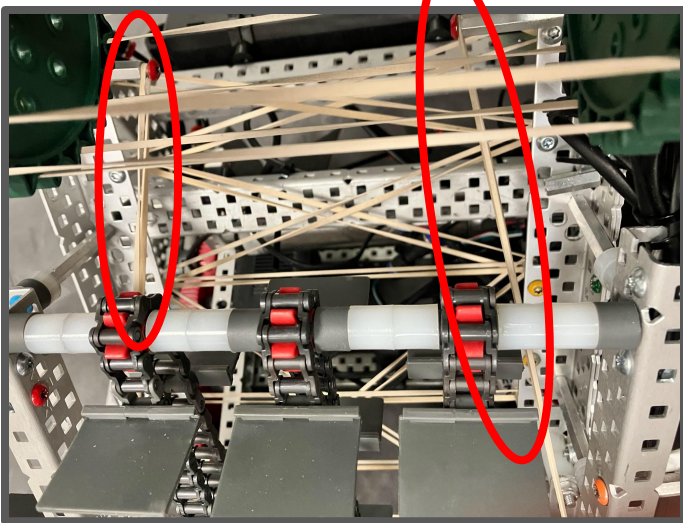
Issues to Address

Without proper guiding structures, blocks frequently flew out of the ramp or jammed near the outtake. The lack of controlled lateral guidance caused blocks to shift unpredictably, reducing scoring consistency and slowing down cycle time.

Changes

To prevent jamming and improve block control, we added two side-mounted rubber bands, each attached to a standoff and secured with a screw. These bands act as flexible guides that keep blocks aligned as they approach the outtake. This ensures that blocks move one at a time, in a straight path, and with far fewer interruptions.

These improvements significantly increase scoring reliability and reduce the risk of jams during high-pressure match situations.



Snow Bots Miller Grove High School Tournament: School Based Tournament Jan.17 - Notes and Reflection

January 17, 2026

Written by: D.M., J.C.
Present: J.C., T.C., D.M., A.B., D.S.

Results

We placed very high in the qualifiers rankings and reached the quarterfinals. We ultimately lost the tournament, but were awarded the design award.

Observations

Competition

At the competition, we noticed a lot more developed robots following a similar design as our Ruiguan Build.

Our Team

With the recent redesigns, our team was able to effectively score blocks, align itself, and actually utilize our autonomous.

Successes

- Placed highly in the qualifier rankings with a win loss ratio of 5-2

Failures

- Autonomous became inconsistent after the first two matches.

Snow Bots Miller Grove High School Tournament: School Based Tournament							
Qualifier #2	Jan 17th at 9:18 AM	4148B	597C	107	42329B	9950R	8
Qualifier #15	Jan 17th at 10:13 AM	42329C	9950T	0	70291A	597C	84
Qualifier #20	Jan 17th at 10:33 AM	597C	70291G	69	41411C	41411A	11
Qualifier #25	Jan 17th at 10:52 AM	597B	597C	30	9950V	41411B	29
Qualifier #36	Jan 17th at 11:35 AM	42786B	42329E	19	597C	42329D	14
Qualifier #44	Jan 17th at 12:48 PM	2750B	2750A	64	2750H	597C	30
Qualifier #50	Jan 17th at 1:17 PM	1394K	597C	11	70291E	30119A	17
QF #2-1	Jan 17th at 3:21 PM	42329A	597F	28	597C	597A	0
R16 #4-1	Jan 17th at 2:46 PM	597C	597A	97	9950T	42329D	0
Rank	8						
WP / AP / SP	10 / 45 / 103						

Viking Robotics: The Battle For Valhalla Tournament: School Based Tournament Jan.31 - Notes and Reflection

January 31, 2026

Written by: D.M.

Present: J.C., T.C., D.M., A.B., D.S.

Results

We placed very high in the rankings along with placing high in the skill rankings. Our team also managed to reached the semifinals, however we ultimately lost the tournament, but were awarded the design award.

Observations

Competition

At the competition, we noticed a lot more developed robots following a similar design as our Ruiguan Build.

Our Team

With the recent redesigns, our team was able to effectively score blocks, align itself, and actually utilize our autonomous.

Successes

- Placed highly in the qualifier rankings with a win loss ratio of 5-2

Failures

- Autonomous was not always consistent and they easily pushed our bot from scoring in the long goal.

Qualifier #5	Jan 31st at 9:45 AM	77711P	597C	121	75160A	8188N	6
Qualifier #10	Jan 31st at 10:10 AM	597F	75160E	37	597C	8188S	73
Qualifier #15	Jan 31st at 10:33 AM	597C	77711S	52	597D	20480H	62
Qualifier #22	Jan 31st at 11:09 AM	75160Y	597C	22	77711J	20480T	67
Qualifier #28	Jan 31st at 11:38 AM	8188F	77711X	3	597C	20480K	47
Qualifier #36	Jan 31st at 1:17 PM	76642T	3691A	14	597C	20480C	89
Qualifier #43	Jan 31st at 1:52 PM	597E	597B	25	597C	41411A	69
QF #2-1	Jan 31st at 3:27 PM	41411B	75160Y	17	597C	597E	86
SF #1-1	Jan 31st at 3:45 PM	20480H	77711P	57	597C	597E	54
R16 #4-1	Jan 31st at 3:04 PM	597C	597E	69	20480T	8188N	17

Rank

5

WP / AP / SP

10 / 60 / 159

Dallas ISD Skyline's 9th Annual VRC Tournament - Notes and Reflection

February 14, 2026

Written by: D.M.

Present: J.C., T.C., D.M., A.B., D.S.

Results

We placed moderately high to comfortably make it to alliance selection. Unfortunately we lost in the quarterfinals.

Observations

Competition

At the competition, we noticed a lot more developed robots following a similar design as our Ruiguan Build.

Our Team

Although our robot could score effectively, our matchload took a while to align to utilize and our ramp elevator was slow.

Successes

- Our robot was able to score and matchload.

Failures

- Although we were able to descore blocks, we were unable to utilize our descore mech to its full potential. The strategy for the game is very dependent on the descore.

Dallas ISD Skyline's 9th Annual VRC Tournament							
Qualifier #1	Feb 14th at 9:20 AM	8514B	597C	12	20480H	99689B	49
Qualifier #14	Feb 14th at 11:29 AM	55577T	597C	68	99689C	3691A	29
Qualifier #22	Feb 14th at 12:29 PM	55577R	20480K	27	597C	67679B	15
Qualifier #31	Feb 14th at 1:08 PM	21063B	8188Z	50	20480J	597C	58
Qualifier #42	Feb 14th at 1:56 PM	8514A	10572B	6	3028G	597C	109
Qualifier #48	Feb 14th at 2:16 PM	597F	20480C	83	17227X	597C	11
QF #1-1	Feb 14th at 4:32 PM	597A	20480K	60	597C	18350A	16
R16 #2-1	Feb 14th at 4:03 PM	17227X	49958B	0	597C	18350A	65
Rank	14						
WP / AP / SP	6 / 30 / 123						

Drivetrain Redesign

February 16 , 2026

Written by: D.M and J.C

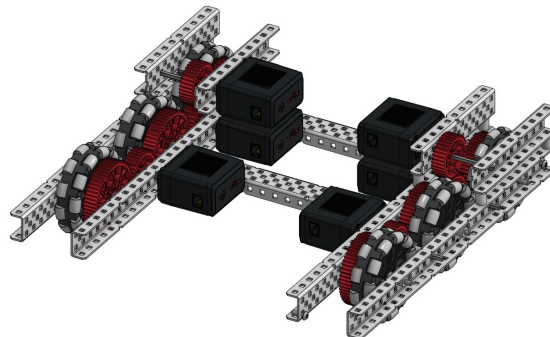
Present: J.C., D.M., A.B., T.C., D.S.

Goals

- *Replace the two center omni wheels from the left and right side of our drivetrain with traction wheels in order to prevent horizontal sliding and pushing.*
- *Lower the center of mass by increasing the weight of the drivetrain to mitigate tipping.*

Previous design

Our original drivetrain was geared for torque which gave us stronger pushing ability and allowed us to hold our ground against other robots. However, using only omni wheels made it easy for use to be pushed around horizontally.



Issues to address

Although we were geared for torque, using only omni wheels made it easy for use to be pushed around horizontally. To mitigate this, we will replace the two center omni wheels from the left and right side of our drivetrain with traction wheels. Another issue to address is tipping. To mitigate tipping, we will lower the center of mass of our robot by moving components down such as the brain and pneumatics tanks to the lower center of the robot. That is, move the pneumatics tanks at the bottom of the drivetrain and place the brain right above the drivetrain.

Drivetrain Redesign Cont.

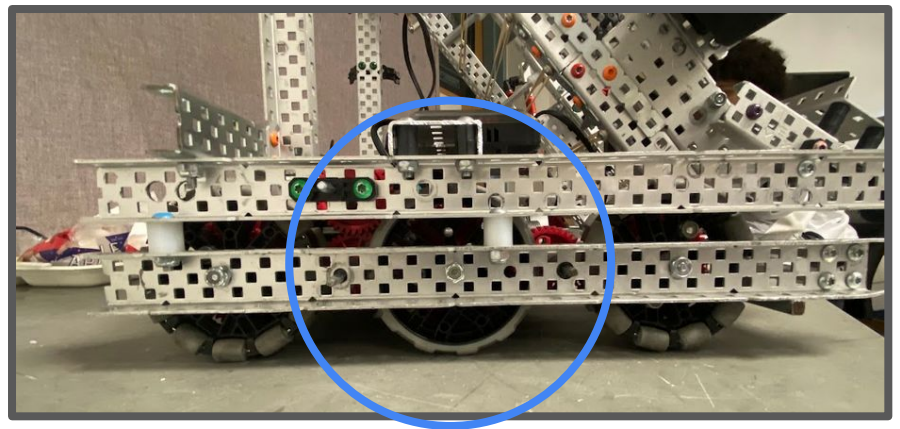
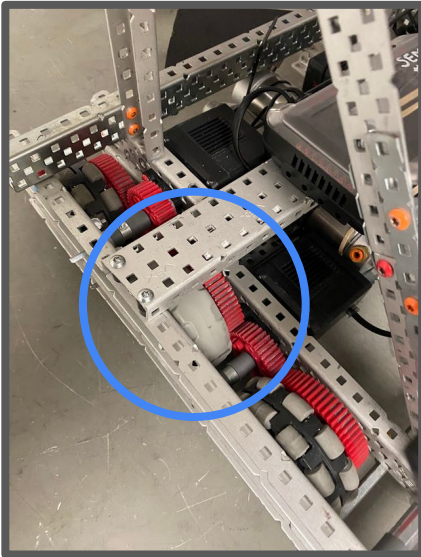
February 16 , 2026

Written by: D.M and J.C

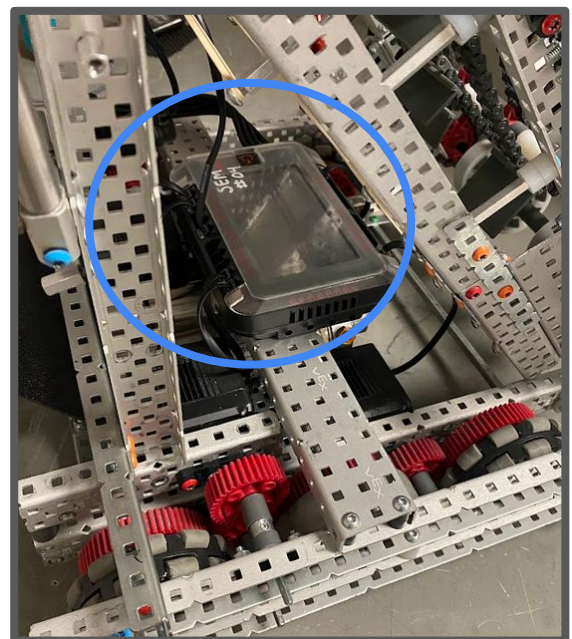
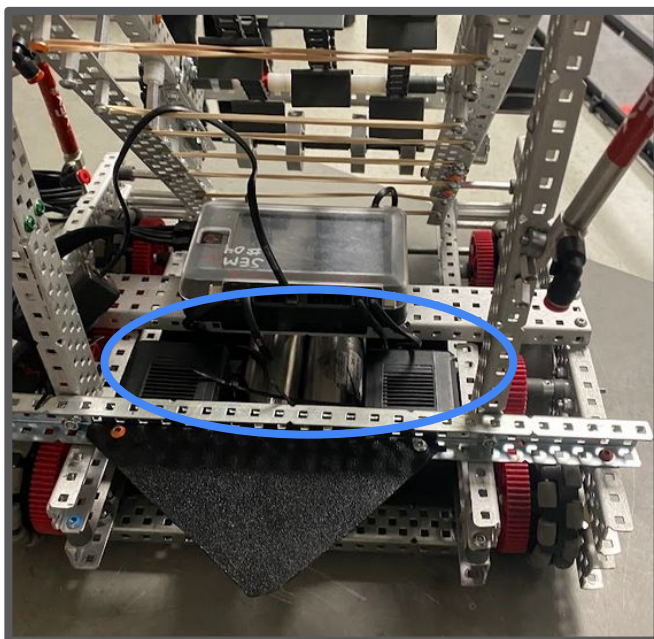
Present: J.C., D.M., A.B., T.C., D.S.

New Design Solution

Swapping the center omniwheels with traction wheels:



Placing tanks at the bottom of the drivetrain and placing the brain right above the drivetrain:



Ramp Elevator Improvement

February 17 , 2026

Written by: D.M.

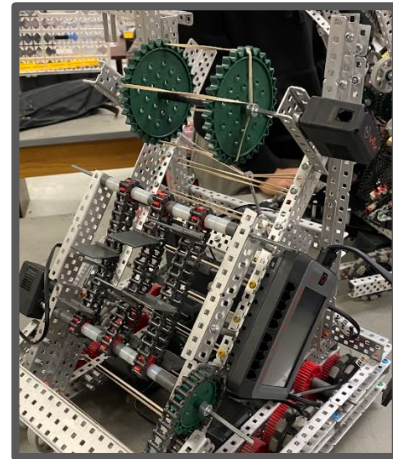
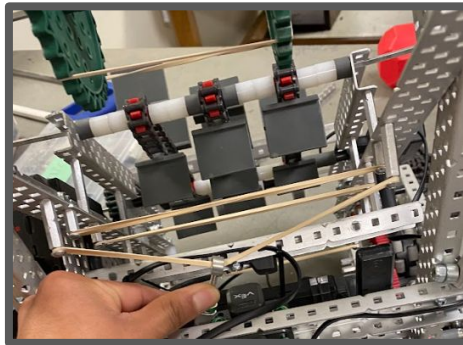
Present: J.C., D.M., A.B., T.C., D.S.

Goals

- *Increase the spacing between the chain flaps and the ramp to create more room for blocks to move smoothly through the Ramp Elevator.*

Previous design

In the earlier design, the spacing between the chain flaps and the rubber-band ramp was very tight.

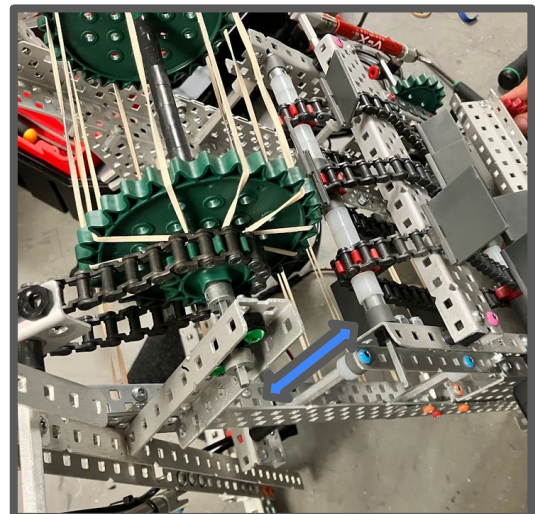


Issues to address

Because the gap was too small, blocks would occasionally get stuck while traveling up the ramp. This slowed down the elevator system and reduced scoring consistency. To prevent these jams, we needed to increase the spacing between the chain flaps and the ramp surface

New Design Solution

Originally, our chain was angled down so the space between the chain and the floor of the ramp became narrower as you went down the ramp. What we did was shift the end of the elevator chain to be at the same level as the top so that the elevator/conveyor had even spacing throughout the ramp. Thus, increasing the spacing between the ramp and the elevator chain.



Ramp Funnel Redesign

February 18 , 2026

Written by: D.M.

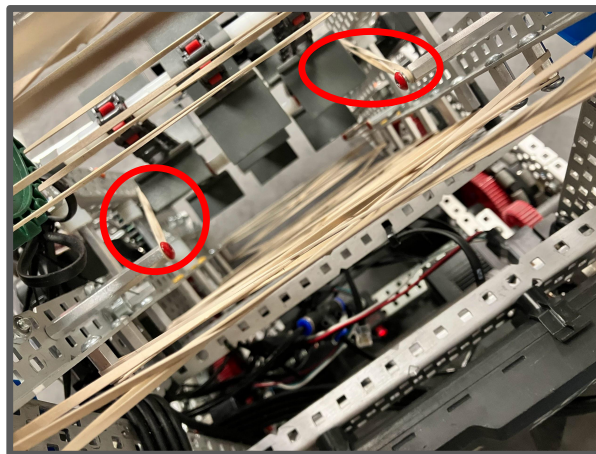
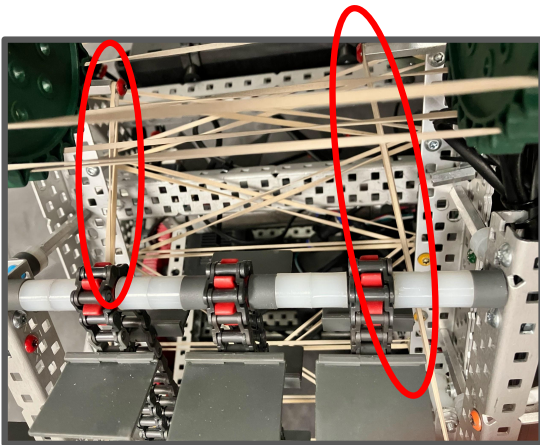
Present: J.C., D.M., A.B., T.C., D.S.

Goals

- Create a self-compressing ABS funnel that improves block flow and increases the speed and smoothness of the Ramp Elevator.

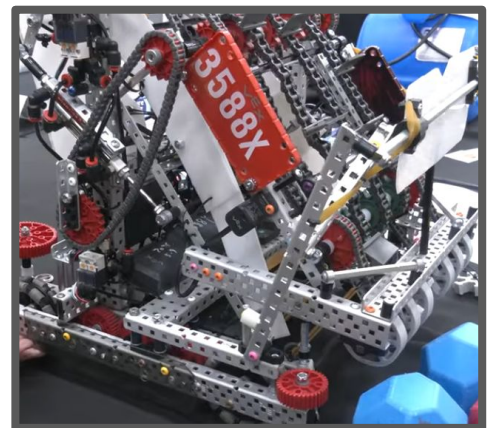
Previous design

Our earlier ramp funnel relied on rubber bands arranged to guide the blocks upward. While this system worked, it did not provide the smoothness or speed we needed. The rubber bands flexed inconsistently, which sometimes slowed block movement or caused minor misalignment.



New Design Solution

We are replacing the rubber-band funnel with a rigid ABS funnel. The new funnel will be wide at the bottom to easily capture incoming blocks and will gradually narrow as it moves upward, compressing the blocks into a controlled, centered path. This tapered shape ensures that blocks enter the elevator consistently and maintain alignment as they travel up the ramp. The ABS material provides a smoother, more predictable surface than rubber bands, reducing friction points and improving overall flow. This design mirrors the proven funnels used by high-performing teams and will significantly increase the speed and reliability of our intake-to-elevator transition.



Credit:
<https://www.youtube.com/watch?v=iKctGfctif8>

Matchload Redesign

February 25, 2026

CWritten by: D.M. and J.C

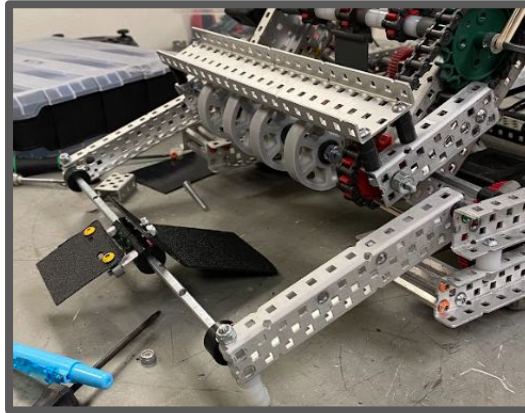
Present: J.C., D.M., A.B., T.C., D.S.

Goals

- *Develop a new match-load mechanism capable of loading blocks from multiple angles to make unloading easier. Add small omni-wheels to allow the match-load to roll smoothly when extended.*

Previous design

Our earlier match-load used two small ABS pieces that were prone to breaking. It also required a very specific angle to function properly, which made match-loading inconsistent and difficult during matches

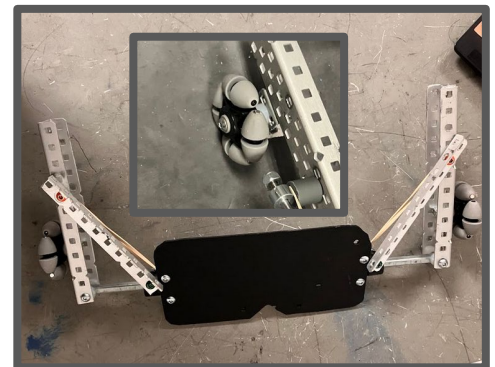


Issues to address

- The previous design lacked omni-wheels, causing it to drag and get stuck on the foam field tiles.
- The ABS pieces were fragile and frequently broke under pressure.
- The mechanism only worked at a narrow angle, limiting how effectively we could unload blocks

New Design Solution

We decided to construct a matchload that is a single piece of abs to reduce the number of abs pieces that could break. This design utilizes rubberbands to bend back into position after match loading. It also utilizes omni wheels to roll above the field foam for better maneuverability.



RoboBusters Blue

SEM

Robotics Engineering

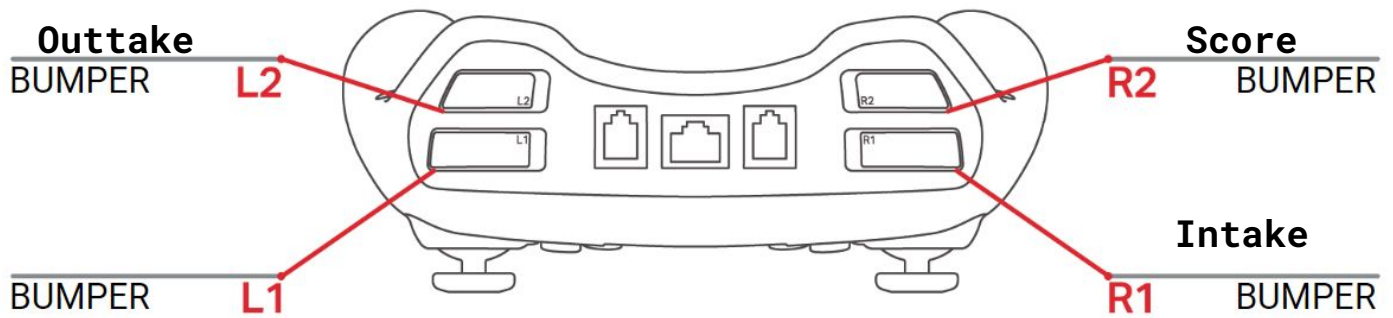
Notebook

Section B - Programming

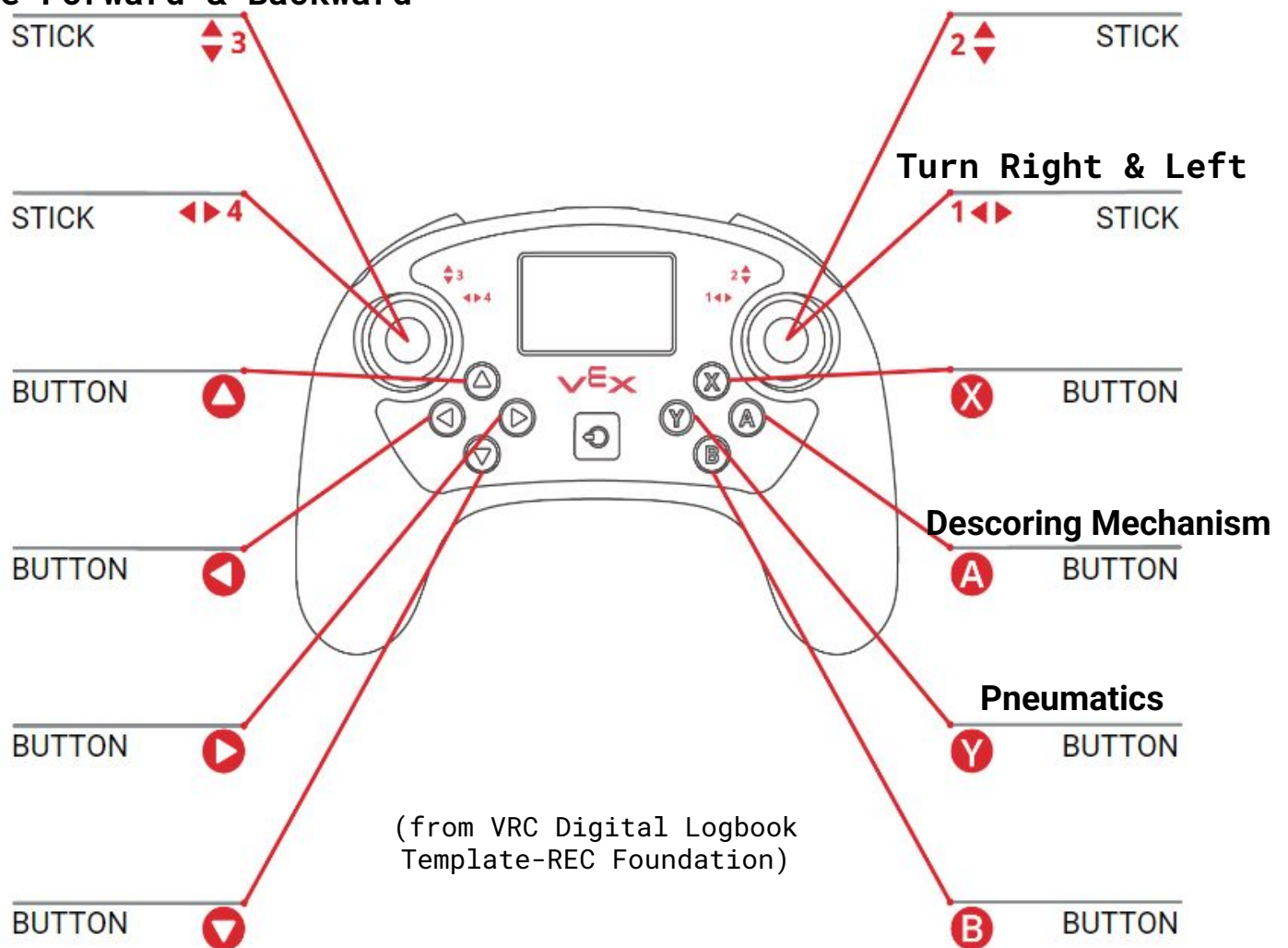
Button Map

Written by: T.C.

Present: T.C., A.B., J.C.



Move Forward & Backward



(from VRC Digital Logbook
Template-REC Foundation)

DATA NORMALIZATION

August 12, 2025

Written by: T.C.

Present: T.C.

PURPOSE (WHY)

Scales sensor data into usable ranges for high-level algorithms.

MECHANISM (HOW)

Pure C utility functions for linear scaling and bounds checking.

IMPLEMENTATION LOGIC

- Portability: Side-effect free code for reusability across subsystems.

SYSTEM Initialization

August 12, 2025

Written by: T.C.

Present: T.C.

PURPOSE (WHY)

Ensures deterministic startup to prevent hardware race conditions.

MECHANISM (HOW)

Sequences hardware: Platform -> Sensors -> Actuators. Uses imu_is_ready() gate.

IMPLEMENTATION LOGIC

- Safety: MOTORS DISABLED for 2s during IMU calibration to prevent bias.
- Timing: Heavy tasks are non-blocking to maintain 10ms loop frequency.

CONSTANTS configuration

August 23, 2025

Written by: T.C.

Present: T.C.

PURPOSE (WHY)

Centralizes 'magic numbers' for quick field-side tuning.

MECHANISM (HOW)

Uses inline constexpr values for physical dims and PID gains.

IMPLEMENTATION LOGIC

- Units: Every constant is documented (m, kg, s) to prevent ambiguity.

DRIVETRAIN: Initialization

August 23, 2025

Written by: T.C.

Present: T.C.

PURPOSE (WHY)

Establishes robot coordinate system and hardware motor state.

MECHANISM (HOW)

Resets internal odometry (Pose {x, y, heading}) and zeroes encoder counts.

IMPLEMENTATION LOGIC

- Standard: All distance math in Meters to avoid conversion errors.
- Error: If encoders are noisy, flags ERROR and disables closed-loop commands.

PID CONTROLLER

August 23,2025

Written by: T.C.

Present: T.C.

PURPOSE (WHY)

Provides smooth corrective control for autonomous by minimizing error over time.

MECHANISM (HOW)

Calculates effort based on Proportional, Integral (accum), and Derivative terms.

IMPLEMENTATION LOGIC

- Anti-windup: Clamps 'integral_accum' to prevent massive overshoots after stalls.
- Safety: Catch $dt=0$ cases to avoid system-crash division errors.

DRIVETRAIN: SETVELOCITY

August 25, 2025

Written by: T.C.

Present: T.C.

PURPOSE (WHY)

Enables precise speed control (m/s) regardless of battery voltage.

MECHANISM (HOW)

Converts linear target (m/s) to Motor RPM using wheel circumference/gearing.

IMPLEMENTATION LOGIC

- Closed-Loop: Uses Feedforward + PID to reach and hold target speed.

DRIVETRAIN: SET OPEN LOOP

August 25, 2025

Written by: T.C.

Present: T.C.

PURPOSE (WHY)

Provides direct operator control without PID interference.

MECHANISM (HOW)

Scales percent (-1.0 to 1.0) directly to motor API units for raw output.

IMPLEMENTATION LOGIC

- Safety: Commands ignored if a safety lock is active (e.g., calibration).

INTAKE SUBSYSTEM

August 25, 2025

Written by: T.C.

Present: T.C.

PURPOSE (WHY)

Manages collection and identifies jams before motor burnout.

MECHANISM (HOW)

Controls rollers with current-sensing jam detection logic.

IMPLEMENTATION LOGIC

- Recovery: If current > threshold, reverses briefly and logs a WARN.

```
if (master.get_digital(E_CONTROLLER_DIGITAL_R1)) {  
    it.intake();  
} else if (master.get_digital(E_CONTROLLER_DIGITAL_R2)) {  
    it.score();  
} else if (master.get_digital(E_CONTROLLER_DIGITAL_L2)) {  
    it.outTake();  
} else {  
    it.stop();  
}
```

Programming: Constants File Setup

September 15, 2025

Written by: T.C., J.C.

Present: T.C., J.C.

Goal

Establish a centralized configuration file to store motor ports, drivetrain geometry, encoder values, and conversion formulas. This ensures consistency across all subsystems and simplifies future tuning.

Program Summary

Method Name	Snippet	Algorithm	Purpose
headingRestrict()	heading = fmod(std::abs(heading), pi*2);	Wraps heading angle within 0–360 degrees	Ensures consistent heading values for autonomous control
distancePerTick	wheelCircumference / (ticks * gearRatio)	Converts encoder ticks to inches	Used for precise movement calculations
inchesPerTick	wheelDiameter * pi / unitsToRevolution	Converts motor units to inches	Used for drive encoder scaling

Sensor Summary

Sensor Name	Sensor Type	Used in Method(s)	Purpose
Motor Encoders	Internal	All movement calculations	Converts motor ticks to physical distance

```
const int8_t fl_p = 11; // Port of Front Left motor
static constexpr unsigned char ml_p = 12; // Port of Middle Left motor
static constexpr unsigned char bl_p = 13; // Port of Back Left motor
static constexpr unsigned char fr_p = 16; // Port of Front Right motor
static constexpr unsigned char mr_p = 17; // Port of Middle Right motor
static constexpr unsigned char br_p = 18; // Port of Back Right motor

static constexpr unsigned char it1_p = 1; // Port of the Elevation mech
static constexpr unsigned char it2_p = 21; // Port of the Elevation mech
static constexpr unsigned char it3_p = 2; // Port of the Elevation mech
```

Programming: DriveTrain Subsystem Implementation Part 1

September 15, 2025

Written by: T.C., J.C.

Present: T.C., J.C.

Goal

- *Build a six-motor drivetrain class with support for arcade and tank drive.*
- *Add autonomous movement and turning methods using encoder feedback and PID control.*

Program Summary

Method Name	Snippet	Algorithm	Purpose
arcadeDrive()	left_g.move(leftY + rightX);	Combines forward and turning input	Enables intuitive driver control
tankDrive()	left_g.move(leftY); right_g.move(rightY);	Independent left/right control	Alternative teleop driving method
turnAngle()	pidController.compute(position)	PID loop based on encoder feedback	Turns robot to a specific angle
turnAngleNew()	move_relative(Direction * ticks, maxRPM);	Relative motor movement	Simplified angle turning using directional input
moveForward()	move_relative(moveAmount, 100);	Direct relative movement	Moves robot forward by motor units

Sensor Summary

Sensor Name	Sensor Type	Used in Method(s)	Purpose
Motor Encoders	Internal	All movement methods	Tracks position for distance and angle control
PID Controller	Software	turnAngle()	Maintains accurate turning using feedback loop

Programming: DriveTrain Subsystem Implementation Part 1

September 15, 2025

Written by: T.C., J.C.

Present: T.C., J.C.

Goal

- *Create a three-motor intake class for elevation and ball control. Implement spin-up, spin-down, and autonomous timed spin*

Program Summary

Method Name	Snippet	Algorithm	Purpose
spinUp()	it_mtr.move(127);	Sets all motors to full forward speed	Activates intake to collect game elements
spinDown()	it_mtr.move(-127);	Sets all motors to full reverse speed	Reverses intake to eject or reposition elements
spinout()	MotorGroup({-it1_p, it2_p, it3_p})	Custom motor direction configuration	Spins intake with alternate motor polarity
autonSpin()	it_mtr.move(-voltage); delay(duration);	Timed voltage-based control	Spins intake during autonomous Autonomous period
stop()	it_mtr.move(0);	Stops all intake motors	Halts intake movement

Sensor Summary

Sensor Name	Sensor Type	Used in Method(s)	Purpose
Motor Encoders	Internal	moveHorizontal, turnAngle	Tracks position for movement and turning

T.C. 9/15/25

Programming: DriveTrain Subsystem Implementation Part 1

September 15, 2025

Written by: T.C., J.C.

Present: T.C., J.C.

Goal

- *Create a three-motor intake class for elevation and ball control. Implement spin-up, spin-down, and autonomous timed spin*

```
private:
    Motor it1_mtr = Motor(it1_p);
    Motor it2_mtr = Motor(it2_p);
    Motor it3_mtr = Motor(it3_p);

    MotorGroup it_mtr = MotorGroup({it1_p, it2_p, -it3_p});
```

```
void stop() {
    it_mtr.move(0);
}

void spinUp() {
    MotorGroup it_mtr = MotorGroup({it1_p, it2_p, -it3_p});
    it_mtr.move(127);
}

void spinDown() {
    MotorGroup it_mtr = MotorGroup({-it1_p, it2_p, -it3_p});
    it_mtr.move(-127);
}
```

```
void spinout() {
    MotorGroup it_mtr = MotorGroup({-it1_p, it2_p, it3_p});
    it_mtr.move(127);
}
```

Programming: DriveTrain Subsystem Implementation Part 2

September 15, 2025

Written by: T.C.
Present: T.C., J.C.

Goal

- *Build a six-motor drivetrain class with support for arcade and tank drive. Add autonomous movement and turning methods using encoder feedback and PID control.*

```
static const double gearRatio = 3.0/5.0;  
static const int encoder_ticks_per_rev = 1800;  
static const double wheelCircumference = 3.25 * pi;  
static const double distancePerTick = wheelCircumference/(encoder_ticks_per_rev*gearRatio);
```

```
static const double pi = 3.14159265358979323846; // pi  
static const double radToDeg = 180/pi;  
static constexpr float wheelDiameter = 3.25f; // Diameter of drive wheels  
  
//drive train auton values-----  
static const float trackwidth = 14.18f;  
static const float kP = 0.5;  
static const float kI = 0.5;  
static const float kD = 0.5;  
static const float radius = trackwidth; // can be found by measuring the width of the drive train
```

T.C. 9/15/25

Programming: Autonomous – Initial Pathing and Intake Part 1

October 20, 2025

Written by: T.C., J.C.

Present: T.C.

Goals

- *Begin building a right-side Autonomous period that scores multiple balls on the tall goal. Focus on intake activation with drive movement and verifying basic pathing.*

Program Summary

Method Name	Snippet	Algorithm	Purpose
moveHorizontal()	dt.moveHorizontal(25, 50);	Moves robot forward	Approaches first ball position
spinUp()	it.spind();	Activates intake	Intake balls
turnAngle()	dt.turnAngle(-20);	Turns robot left	Aligns with balls in on corner of mat

Sensor Summary

Sensor Name	Sensor Type	Used in Method(s)	Purpose
Motor Encoders	Internal	moveHorizontal, turnAngle	Tracks position for movement and turning

```
void autonomous() {  
    dt.moveHorizontal(25,50 );// move forward for 1 second at 100% speed  
    it.spinUp();  
    dt.turnAngle(-20);  
}
```

Programming: Autonomous – ball and goal alignment

October 20, 2025

Written by: T.C., J.C.

Present: T.C.

Goals

- *Begin building a right-side Autonomous period that scores multiple balls on the tall goal. Focus on intake activation with drive movement and verifying basic pathing.*

Program Summary

Method Name	Snippet	Algorithm	Purpose
turnAngle()	dt.turnAngle(-30);	Turns robot left	Aligns with final ball
moveHorizontal()	dt.moveHorizontal(16.5, 70);	Moves robot forward	Approaches tall goal
spinDown()	it.spinDown();	Reverses intake	Scores balls into goal

Sensor Summary

Sensor Name	Sensor Type	Used in Method(s)	Purpose
Motor Encoders	Internal	moveHorizontal, turnAngle	Tracks position for movement and turning

Notes

The full Autonomous period was tested successfully. The robot scored four balls on the tall goal from the right side. Intake reversal timing was critical for the final ball. Minor drift during the last turn was noted but didn't affect scoring. Autonomous period is competition-ready with small refinements.

Programming: Autonomous – Initial Pathing and Intake Part 2

October 21, 2025

Written by: T.C., J.C.

Present: T.C.

Goals

Begin building a right-side Autonomous period that scores multiple balls on the tall goal. Focus on intake activation with drive movement and verifying basic pathing.

Program Summary

Method Name	Snippet	Algorithm	Purpose
moveHorizontal()	dt.moveHorizontal(10, 40);	Short forward movement	Positions robot for second ball
turnAngle()	dt.turnAngle(40);	Turns robot right	Aligns Robot to back up and become inline with tall goal
moveHorizontal()	dt.moveHorizontal(-28.5, 80);	Moves robot backward	Retreats to scoring position
stop()	it.stop();	Stops intake	Ends collection phase

Sensor Summary

Sensor Name	Sensor Type	Used in Method(s)	Purpose
Motor Encoders	Internal	moveHorizontal, turnAngle	Tracks position for movement and turning

DeScore

October 29, 2025

Written by: T.C.

Present: T.C.

PURPOSE (WHY)

Reliable element manipulation via position-controlled clamping.

MECHANISM (HOW)

Uses limit switches to zero position and encoders for precise positioning.

IMPLEMENTATION LOGIC

- Calibration: Zeros at startup using limit switches for absolute accuracy.

```
if (master.get_digital_new_press(E_CONTROLLER_DIGITAL_A)) {  
    deScoreState = !deScoreState;  
    mbc.changedeScore(deScoreState);  
}
```

MATCHLOAD CYCLE

November 25, 2025

Written by: T.C.

Present: T.C.

PURPOSE (WHY)

Coordinates subsystems for fast, deterministic piece loading.

MECHANISM (HOW)

A state machine sequences the Intake, Indexer, and Feeder via sensor gates.

IMPLEMENTATION LOGIC

- Flow: IDLE -> INTAKING -> INDEXING -> COMPLETE.

```
if (master.get_digital_new_press(E_CONTROLLER_DIGITAL_B)) {  
    loadState = !loadState;  
    ml.changeLoadState(loadState);  
}
```

IMU WRAPPER

January, 6, 2026

Written by: T.C.

Present: T.C.

PURPOSE (WHY)

Corrects gyro bias and drift for a stable, fused heading.

MECHANISM (HOW)

Integrates gyro data while applying an Alpha-smoothing filter (~ 0.15) to reduce noise.

IMPLEMENTATION LOGIC

- Wrapping: Normalizes output to $[-180, 180)$ to prevent control loop jumps.
- Shock: Ignores accelerometer spikes during calibration phases.

COLOR SORT LOGIC

January,16,2026

Written by: T.C.

Present: T.C.

PURPOSE (WHY)

Automates game piece routing for 'Pushback' scoring.

MECHANISM (HOW)

Uses a circular buffer to average samples, filtering out lighting interference.

IMPLEMENTATION LOGIC

- Hysteresis: Prevents actuator 'chatter' near detection thresholds.
- Non-Blocking: State machine runs in background to keep drive loops fast.

Absolute Position Turning

March 6, 2026

Written by: T.C.

Present: T.C.

The Goal:

Moving from "Relative" turns (turning a specific amount from where you are) to "Absolute" turns (facing a fixed direction on the field).

The Why (The Problem):

In a relative system, if your robot starts 5 degrees crooked or gets bumped, every subsequent turn is off by that same amount. Errors compound. In an absolute system, the field is a fixed map. 0° is always "North." If you tell the robot to face 90°, it doesn't matter if it started at 0° or 10°—it will end up at exactly 90°.

The How (The Mechanism):

The robot uses the **Inertial Measurement Unit (IMU)** to track its "Global Heading."

1. **Calculate Error:** $\text{Error} = \text{Target Angle} - \text{Current Angle}$.
2. **Shortest Path Logic:** The robot must decide whether to turn clockwise or counter-clockwise. For example, to go from 350° to 10°, it's smarter to turn +20° than -340°.
3. **Correction:** The error is fed into a PID loop that slows the motors down as the robot approaches the target angle to prevent overshooting.

On-Screen Autonomous Selector

March 6, 2026

Written by: T.C.

Present: T.C.

Goal:

A user interface on the V5 Brain that allows you to swap programs without touching a laptop.

The Why:

At a competition, your strategy changes based on your alliance partner. If they can do the "AWP" (Autonomous Win Point) on the left, you need to be able to instantly switch to the right-side routine.

The How:

The Brain's touchscreen acts as a set of digital toggle switches.

1. **State Storage:** A variable (like `auton_count`) is stored in the code.
2. **User Input:** Every time you tap a specific area of the screen, the `auton_count` increases (1, 2, 3...).
3. **Visual Feedback:** The screen prints the name of the routine (e.g., "Red Side - 4 Rings") so you have visual confirmation of what is about to run.
4. **Execution:** When the competition switch hits "Autonomous," a `switch` statement checks the value of `auton_count` and runs the corresponding function.

GitHub Repositories

Written by: T.C.

Present: T.C.

Multiple repositories used

Current : https://github.com/Tyc599/597C_2025-main-real

Previous: https://github.com/Tyc599/597C_2025-main
https://github.com/Tyc599/597C_2025

Live Code Updates:

https://github.com/Tyc599/597C_2025-main-real/tree/main/docs/logbook